



Creating a Cancer Patient-facing Question Answering System using LLMs

by

MaoLin He

Student ID:1350038

Supervisor: A/Prof Brian Chapman

A thesis submitted in total fulfillment for the
degree of Master of Information technology

in the

Master of Information Technology

School of Computing and Information Systems

THE UNIVERSITY OF MELBOURNE

August 2024

THE UNIVERSITY OF MELBOURNE

Abstract

Master of Information Technology
School of Computing and Information Systems

Master of Information technology

by MaoLin He
Student ID:1350038

Supervisor: A/Prof Brian Chapman

Large Language Models (LLMs) have revolutionized question-answering systems but face “hallucination” in knowledge-intensive domains like medical oncology. This study addresses the research question: “In Cancer Patient Question Answering (CPQA) systems based on Retrieval-Augmented Generation (RAG), how can we get the right information passed to the right LLM to generate output in the right form?” We propose novel techniques to enhance CPQA performance and efficiency. First, our Single-Step Comprehensive Retrieval With Evidence-Processing (SCREP) integrates optimized query pipelines and a novel Hierarchical Evidence-Processing (Hier-EP) method, which obtains high-quality evidence within comprehensive retrieval results from diverse medical sources and passes evidence to prompt with adaptive granularity control. We demonstrate SCREP’s effectiveness across various LLMs and datasets, significantly improving CPQA performance. For example, only SCREP elevates LLaMA-3-8B’s average response accuracy above the original Claude-3-haiku. Through the comparison experiment, we find that smaller, open-source LLaMA-3-8B can balance performance and efficiency. Then, we analyzed the limitations of using LLM’s self-assessment for negative rejection, and developed a Comprehensibility and Verifiability Booster (CVB) to ensure patient-friendly outputs with improved readability and source tracking. Third, we present Multi-Criteria Adaptive RAG (MCA-RAG), a novel approach that dynamically selects the most suitable retrieval approaches based on necessary criteria. MCA-RAG incorporates a novel Dual-LLM Collaborative React-based Multi-Step Retrieval (DCRMR) for complex queries, leveraging the strengths of two LLMs to reduce latency and improve retrieval quality. Our experimental results demonstrate that MCA-RAG elevates LLaMA-3-8B’s average response accuracy above the original GPT-4 while maintaining computational efficiency. This research contributes to the development of more accurate, reliable, and patient-centered cancer question-answering systems.

Declaration of Authorship

I certify that:

- this thesis does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any university; and that to the best of my knowledge and belief it does not contain any material previously published or written by another person where due reference is not made in the text.
- where necessary I have received clearance for this research from the University's Ethics Committee (Approval Number) and have submitted all required data to the School
- The thesis is fewer than 8000 words in length (excluding text in images, tables, bibliographies and appendices).

Signed:

Date:

Acknowledgements

My grandfather was a cancer patient, which is why I chose to create a cancer consumer question-answering system based on large language models for my CIS project. I sincerely hope to contribute something meaningful to cancer patients and their families, like mine. As an ordinary person searching for cancer-related information online, I understand the frustration of encountering conflicting information and the anxiety of possibly missing crucial details. This was why, during a presentation, when Professor Brian asked if I had to choose between online NCBI sources and books as grounding sources, I chose the online NCBI sources. Despite knowing Professor Brian's intent to have me elaborate on which resource is more important, my experience as a family member of a cancer patient led me to prioritize the more comprehensive and accurate online sources. This heartfelt desire has been a driving force behind my continuous learning.

I would like to express my sincere gratitude to my supervisors; Professor Brian for his motivation, guidance and immense knowledge. He have been a pillar of support all along and the best supervisors I could have ever hoped for. Thanks to my supervisors, my initial spark for research is now turning into a big flame. I am actually grateful for this semester's 50 credits of research-related courses. First, there is the change in identity: in the past, I was merely a consumer of knowledge. However, research requires more critical thinking, quickly identifying valuable literature. It involves considering how I can improve and become a producer of knowledge, adding to the existing body of work. Then, there is the enhancement of self-awareness. I realized that I excel in and enjoy the process of reading literature, summarizing, analyzing, and identifying areas for improvement and solution ideas. Designing experiments and analyzing results to confirm hypotheses or discover unusual findings brings me satisfaction. I also discovered that I am prone to self-doubt and indecision without clear research goals and directions, which can interfere with my progress.

The world seldom aligns with our expectations, and you never know what tomorrow holds. I want to commemorate my grandfather in my thesis, ensuring that he is remembered and honored. He and my grandmother were both wonderful people who loved me dearly and supported my dreams and my life. We spent a lot of time together. Although they have passed away, I will remember and love them forever and ensure they are remembered through my papers. We will meet again someday. Even if my work only provides a little help to those suffering from illness and their families like mine, I believe they would be proud of it. I have realized that you never know what tomorrow holds, so you must seize every moment and love those around you. Make every day count.

Contents

Abstract	i
Declaration of Authorship	ii
Acknowledgements	iii
List of Figures	vi
List of Tables	vii
1 Introduction	1
2 Literature review	6
2.1 Single-Step Comprehensive Retrieval With Evidence-Processing	6
2.1.1 Query Pipeline Integration	6
2.1.2 Evidence-Processing	7
2.2 Generation Customization	8
2.3 Non-single-step Retrieval	9
3 Data Collection and Analysis	11
3.1 Data Collection	11
3.2 Analysis Method	12
4 Single-Step Comprehensive Retrieval With Evidence-Processing	13
4.1 Introduction	13
4.2 Method	14
4.2.1 Query Pipeline integration	14
4.2.2 Hierarchical Evidence-Processing (Hier-EP)	16
4.3 Experimental Results and Analysis	18
4.3.1 Query Pipeline integration	18
4.3.2 Hierarchical Evidence-Processing(Hier-EP)	19
5 Generation Customization	22
5.1 Introduction	22
5.2 Method	22
5.2.1 LLM Generator Selection in CPQA	22
5.2.2 Response Adjustment	23

5.3	Experimental Results and Analysis	24
5.3.1	LLM Generator Selection in CPQA	24
5.3.2	Response Adjustment	26
6	Multi-Criteria Adaptive RAG	29
6.1	Introduction	29
6.2	Method	29
6.3	Experimental Results and Analysis	32
7	Conclusion	34
A	Appendix	38
A.1	A glossary of acronyms	38
A.2	Introduction of Original Medical QA dataset	39
A.3	More details(prompt contents) in MCA-RAG	39
	Bibliography	41

List of Figures

1.1	An example of LLM’s hallucinating, where LLMs generate coherent but unfaithful or nonsensical responses [1, 2]. ADW cancer Gene Editing (a made-up treatment name) in the question prompts ChatGPT into hallucination	1
1.2	The Framework comparison between COMP90005 and using LLM alone .	2
1.3	The simplified Framework of MCA-RAG, details see appendix	4
3.1	Introduction of CMMQA and CMCQA	11
4.1	Framework of SCREP	13
4.2	Left: Reciprocal Ranked Fusion (RRF) [3], Right: Relative Score Fusion .	15
4.3	Utilizing E-Utilities’ built-in translator for mapping original terms to their corresponding MeSH terms, synonyms, and lexical variants	17
4.4	Ratio of retrieved evidence from three biomedical corpora using the best method in this section. In subfigure, left: in top-6 evidence, right: in top-12 evidence.	19
4.5	Newer or highly cited studies’ positive impact on QA performance [4] . .	20
4.6	Parameter’s impact on efficiency and accuracy	21
5.1	Evaluation process	23
5.2	Response Accuracy Comparison	25
5.3	Impacts on multiple performance factors and generated time of SCREP .	26
5.4	Case study of RAG with Comprehensibility and Verifiability Boosters (CVB), hallucination phenomena exists in the textual responses	28
6.1	MCA-RAG integrates Knowledge-awareness, Self-awareness and Complexity-awareness: For knowledge-awareness, since CPQA provides patients with medical or emotional counseling, we use a few-shot prompt [5] to determine medical-related queries that need factual knowledge. Considering the importance of accuracy in oncology QA, retrieval is bypassed only when both the LLM prior judgment and the self-awareness classifier deem the given question answerable using the LLM’s internal knowledge. For queries classified as complex, we will use DCRMR.	30
6.2	Upper: Original (Native) ReAct-based Multi-Step Retrieval. Down: Dual-LLM Collaborative ReAct-based Multi-Step Retrieval (DCRMR)	32
7.1	Cancer-related text consumer QA dataset (CTCQA)	36
A.1	The detail Framework of MCA-RAG, this is svg file, suggest zoom in . . .	40

List of Tables

2.1	Answering accuracy of LLMs VS Human [6]	8
4.1	Data Source Statistics, FAISS Index refer to the FAISS index size corresponding to the filtered text(PubMed abstracts in recent ten years, cancer-related segments from NLM books and Textbook)	14
4.2	Experiment Results(ACC); Online NCBI sources: NLM books: Textbook, ratio set 1 is 1:1:1, ratio set 2 is 6:2:1 (discussed in section 4.2.1), the total count of the retrieved document is 72	18
4.3	Performance(ACC) of different methods, experiment default parameters setting: within top-12 evidence, similarity threshold = 0.5 , compression rate = 1	19
4.4	Details of response accuracy in different datasets when changing parameters of Hier-EP(upper-bound, total count) function	21
5.1	Prominent LLMs and their characteristics	22
5.2	Generation accuracy of LLMs+COT on the Filtered Cancer QA Dataset.	25
5.3	Comparison performance, Standard RAG: document retrieved pass to LLM directly, RAG-COMP90005: using best query pipeline in COMP90005	26
5.4	Rejections by Prior Judgment (Prior-J) and Posterior Judgment (Post-J). CR: Correct Rejections, TR: Total Rejections, SR: supposed rejections. In this table, Recall: recall of negative rejection using prior judgment	27
6.1	Effect of criteria in MCA-RAG. MCA-RAG[no DCRM] refers to not using complexity-awareness, and the latter two refer to not using any criterion	32
6.2	The average time per query and the percentage of the predicted labels from the classifiers over all samples	33
7.1	Criteria in MCA-RAG are carefully selected (e.g., Time Awareness and intent awareness are excluded due to domain -specific considerations in section 6.2)	35

Chapter 1

Introduction

Large Language Models (LLMs) like GPT-4 [7], Gemini [8], Claude [9], Mistral [10] and LLaMA [11] have revolutionized the field of question-answering (QA), showcasing exceptional capabilities [12]. However, LLMs’ knowledge relies on their parametric memory, and thus may lack domain-specific or up-to-date knowledge (like medicine information after knowledge cutoff). This limitation can lead to “hallucination” (Figure 1.1). Hallucination is particularly critical in domains requiring high accuracy, reliability, and currency of information, like finance [13] and medicine [14]. Due to hallucination issues, LLMs excel as generators rather than knowledge databases in medical QA systems, where questions are knowledge-intensive [15].

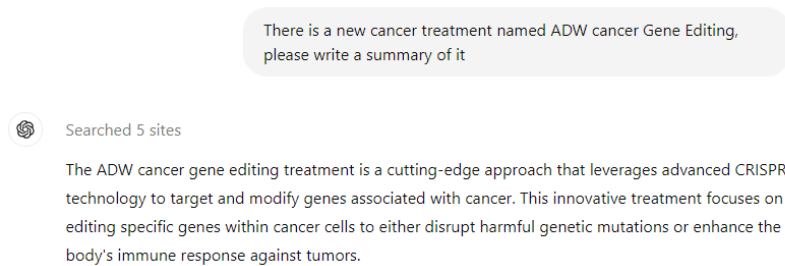


FIGURE 1.1: An example of LLM’s hallucinating, where LLMs generate coherent but unfaithful or nonsensical responses [1, 2]. ADW cancer Gene Editing (a made-up treatment name) in the question prompts ChatGPT into hallucination

Retrieval-Augmented Generation (RAG) [16, 17] integrates external knowledge sources with additional retrieval modules to mitigate hallucination, and has demonstrated enhanced performance in knowledge-intensive tasks [18]. Therefore, we developed a RAG Cancer Patient Question Answering System (RAG-CPQA). Previous studies have shown using RAG improves the accuracy and reliability of medical QA systems [19, 20]. We

have optimized query pipeline components (Figure 1.2) in COMP90005. However, a crucial research question remains: “In RAG-CPQA, how can we get the **right information** passed to the **right LLM** to generate output in the **right form**?”

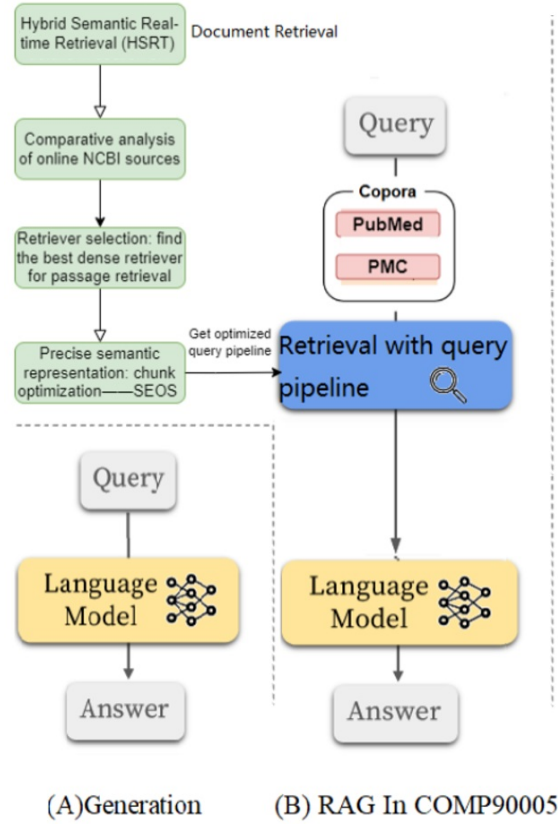


FIGURE 1.2: The Framework comparison between COMP90005 and using LLM alone

Right information means **high-quality evidence** in **comprehensive retrieval results**. Most studies solely use PubMed [4, 21, 22], a reliable, publicly accessible literature resource [23]. Since PubMed only offers literature abstracts, our study in COMP90005 incorporated PMC (PubMed Central) [24] full-text reviews. National Center for Biotechnology Information (NCBI) [25] online sources (PubMed and PMC) focus on biomedical research, which may be inadequate for patients’ needs, and other resources, such as clinical guidelines (e.g. Startpearls [26] in NLM books) may be important. Additionally, passage retrieval in COMP90005 relies on a single retrieval method (dense retrieval [27–29] or BM25 [30]) like most studies [31, 32], which could potentially hinder comprehensive retrieval results. Furthermore, Standard RAG directly appending retrieved information to LLM’s prompt may introduce noise, potentially diminishing response correctness [33], known as the “distraction phenomenon” [34]. Many studies on evidence-processing use relevance as the sole quality factor for evidence-filtering and ignore the text granularity passed to the prompt [35, 36]. However, the medical oncology literature is dynamic [37], has high volume [38], and has varying quality [4], requiring a nuanced approach to

evidence retrieval and processing. Therefore, we explore a novel technique, Single-Step Comprehensive Retrieval With Evidence-Processing (SCREP), and demonstrate that it improves performance across different LLMs and datasets.

Existing medical QA systems [39, 40] focus on clinicians users [41, 42], overlooking patient needs like data privacy and response adjustment. While large-scale models like GPT or Claude demonstrate impressive performance, they raise privacy and security concerns in healthcare [43]. CPQA customization for patients should also consider rejecting potentially unreliable answers (negative rejection) [44] and improving response verifiability. Moreover, the linguistic gap between consumer language and cancer medical terminology may cause CPQA to fail to retrieve information and meet patients’ needs for information comprehensibility [45]. Considering original consumer questions contain orthographic and grammatical errors and vary in length and complexity [46–51], we have employed an LLM as “translator” for two-step query-document alignment process in COMP90005. Firstly, the “translator” uses traditional text normalization processes (like lowercase, tokenization) [52] after errors correction, to get questions with standardized language. Then, processed questions are translated into Boolean expressions, and we have proven that using PubMed’s Boolean operators [53] can improve real-time document retrieval via E-Utilities [54] in COMP90005. To select the **right LLM**, we systematically compared the performance and efficiency of LLMs using different methods on multiple datasets. . Moreover, we analyzed two baselines for negative rejection: 1) using prior confidence in answering a question and 2) using posterior confidence in the generated answer. Finally, we develop a Comprehensibility and Verifiability Booster (CVB) to ensure that providing output in the **right form**.

Recognizing that a one-size-fits-all retrieval approach is suboptimal [31], active retrieval methods consider when to retrieve, but most only consider some criteria of retrieval timing and overlook complex queries [55, 56]. Furthermore, some cancer-related questions often involve intricate relationships between multiple medical entities [57], necessitating multi-step reasoning and retrieval [58]. However, multi-step reasoning and retrieval may introduce challenges like high latency and cascading errors [59]. Considering the diverse needs of queries, we propose a Multi-Criteria Adaptive RAG (MCA-RAG) to judge when and how to retrieve using necessary criteria (knowledge-awareness, self-awareness, and complexity-awareness). For complex queries, we provided a novel Dual-LLM Collaborative React-based Multi-Step Retrieval (DCRMR) to MCA-RAG (Figure 1.3). Finally, we explore the implementation of negative rejection in MCA-RAG.

Our contributions and findings are the following:

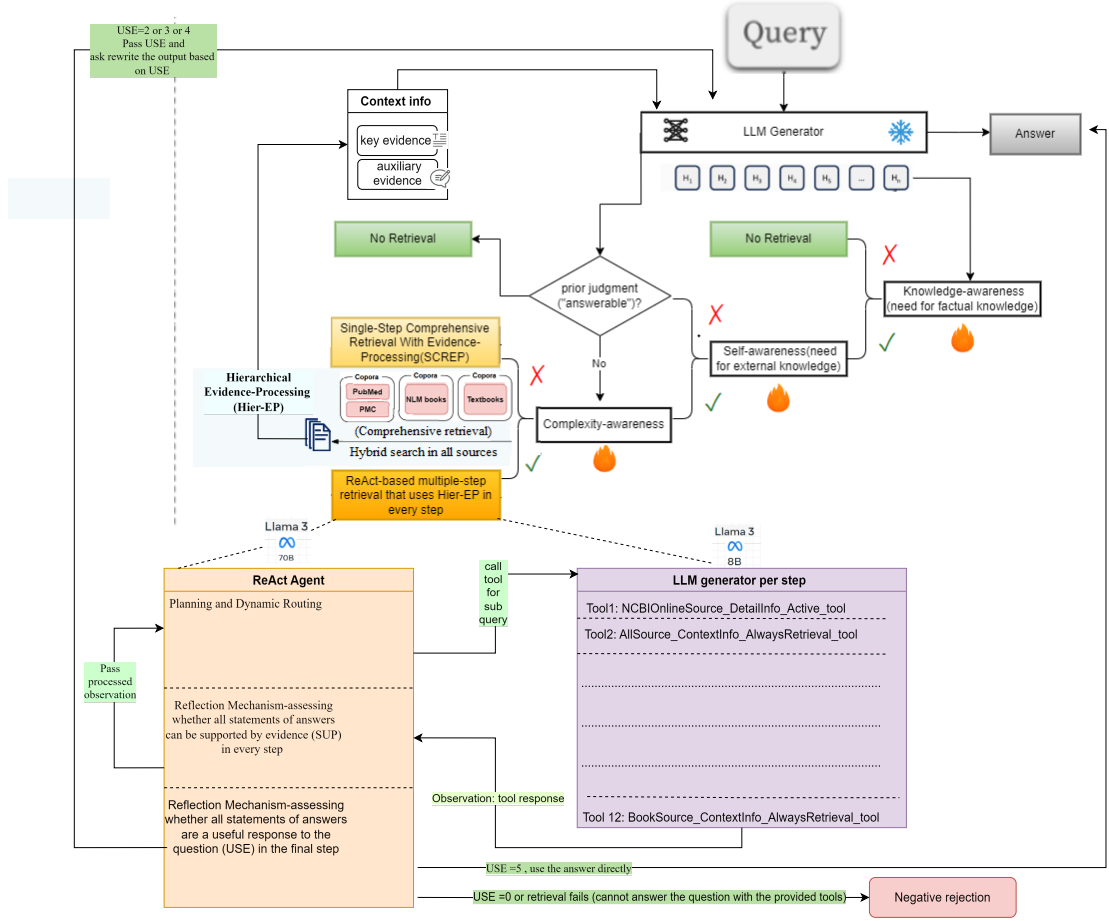


FIGURE 1.3: The simplified Framework of MCA-RAG, details see appendix

- **Single-Step Comprehensive Retrieval With Evidence-Processing (SCREP):** For **comprehensive retrieval results**, we integrated query pipelines and use hybrid search. We found that 1) queries vary in source preference, and 2) NLM books and textbooks offer higher information density than NCBI online sources. For **high-quality evidence**, our novel Hierarchical Evidence-Processing (Hier-EP) is the first to integrate a comprehensive filtering mechanism with adaptive granularity control. By K-means clustering using multiple quality factors (relevance, length, year of publication, number of citations, and data source), whose weights are obtained via automatic tuning, Hier-EP ensures high-quality information is selected considering the relative importance of multiple quality factors. Furthermore, Hier-EP minimizes distraction and focuses on the most relevant information by adaptive granularity control (e.g., only use the most related sentence per auxiliary evidence). We evaluate the performance and efficiency of Hier-EP (upper-bound, total count) concerning the different numbers of total evidence (including key and auxiliary evidence) and upper-bound of key evidence. We find that Hier-EP function parameters should be set based on query characteristics.

- We systematically compare LLMs performance (response accuracy, self-awareness accuracy, and reflection accuracy) and efficiency using different methods on various datasets to select the **right LLM**, showing our SCREP’ generalization and effectiveness (e.g., elevating LLaMA-3-8B’s average response accuracy above the original Claude-3-haiku). Our Comprehensibility and Verifiability Booster (CVB) incorporates source tracking, citations, readability adjustment and fluency optimization, and medical Terminology Simplification to provide generated output in **right form**. Our qualitative analysis of generated output reveals a disagreement between the “step-by-step thinking” analysis and the source text.
- Our novel Multi-Criteria Adaptive RAG (MCA-RAG) is the first work to propose medical QA criteria to adaptively select the most suitable retrieval approaches to obtain **right information** for each query. A key component of MCA-RAG is the Dual-LLM Collaborative React-based Multi-Step Retrieval (DCRMR); DCRMR leverages the strengths of two LLMs to reduce latency and improve retrieval quality. Specifically, LLaMA-3-8B generates per-step responses, while LLaMA-3-70B serves as the ReAct agent, handling high-level reasoning and decision-making tasks such as planning, routing, and reflection. Our MCA-RAG is highly effective and efficient, balancing the response performance (e.g., elevating LLaMA-3-8B’s average response accuracy above the original gpt4) and efficiency.

Chapter 2

Literature review

2.1 Single-Step Comprehensive Retrieval With Evidence-Processing

The single-step RAG system not only includes optimizing the query pipeline (which we did in COMP90005) but also involves query pipeline integration to obtain comprehensive retrieval results, then processing evidence after retrieval to get the right information.

2.1.1 Query Pipeline Integration

Expanding Sources Cancer-related queries necessitate knowledge from biomedical research, medicine, and clinical domains [43], which have different source preferences. In COMP90005, **Hybrid Semantic Real-time document reTrieval** (HSRT) combines MedCPT-based semantic search and E-Utilities [54] to get the NCBI online sources like PubMed abstracts [23] and PMC full-text articles [24], providing valuable biomedical research insights [4, 21, 22]. For clinical queries, medical textbooks and clinical guidelines effectively deliver reliable answers [60], and integrating these sources into LLMs can significantly enhance response quality [61]. In the real world, medical professionals rely on biomedical research literature (the latest findings), medical textbooks (foundational knowledge), and clinical guidelines (authoritative treatment recommendations) for comprehensive learning [62]. This is because cancer-related questions may require insights from diverse sources. Thus, clinical guidelines and textbooks should also be integrated into CPQA to ensure comprehensive and reliable responses.

Single-Step Retrieval Strategies Hybrid search and router search are ensemble retrieval strategies using different retrievers to extract information from distinct sources.

Hybrid search concurrently uses different retrieval methods and sources, and diverse search results which are aggregated using weighted averages [63] or Reciprocal Rank Fusion (RRF) [3] to produce more robust outcomes or rankers to re-sort the search results [64]. This approach improves ranking accuracy but incurs significant computational costs and may introduce noise. The router method uses an LLM as a decision-maker to choose retrieval methods and databases strategically [65]. This will incur additional decision-making costs but may provide a more targeted and efficient retrieval process. However, the effectiveness depends on the design and optimization of the Route model decision function [66]. There is a pressing need to tailor these ensemble retrieval strategies to the medical text used in this study to achieve superior retrieval performance.

2.1.2 Evidence-Processing

Native RAG concatenates the query with the original retrieved evidence in prompt[67]. However, without evidence-processing, the process may result in information overload or exclude evidence context [44], decreasing generated answer quality.

Evidence lacking context can be addressed by the ‘small to big’ principle: retrieving finer content while returning richer context [68]. For instance, Sentence Window Splitter returns the retrieved sentences-group chunk and a window of surrounding sentences [69]. The Semantic Enhanced Overlap Segmentation (SEOS) method developed in COMP90005, which outperforms Sentence Window Splitter by integrating semantic-based splitting and embedding model impacts, also can find context around the retrieved chunk because it establishes inter-chunk relationships via chunk overlap. This means we can use the ‘small to big’ principle when using the SEOS as a splitter. However, the ‘small to big’ principle may introduce noise, affecting response correctness due to the “distraction phenomenon.” Thus, judicious application of the ‘small to big’ principle is crucial.

Information overload means lengthy inputs, which most LLMs struggle to process, slowing down the generation process even using capable models [68]. Lengthy inputs may also include more irrelevant data, causing the model to disregard its own knowledge, even when it has enough internal information to answer [70]. Moreover, too much relevant content can dilute the focus on critical details [71]. One solution is compressing retrieved content, as demonstrated by RECOMP [72], which trains a compressor that condenses documents into concise summaries, and FiD-Light [73], which will compress vectors of retrieved content. Compression reduces latency, but irrelevant data might remain. An alternative is evidence filtering, retaining only content that meets quality or relevance criteria. The simplest way is only using the most relevant evidence; recent

studies suggest that using top-k evidence instead of top-1 can improve performance but may influence RAG system efficiency [74]. Moreover, this fixing top-k increases hyperparameters and lacks flexibility [75]. More sophisticated approaches include adaptive top-k selection. For instance, Self-RAG [55] introduces a self-reflection mechanism that filters out irrelevant content by evaluating retrieved passages for relevance, supportiveness, and utility. However, it is expensive to use LLM to filter adaptive top-k. Further refinement of evidence filtering is token-level elimination. FiD-TF [35] and FILCO [36] focus on removing irrelevant or redundant tokens from retrieved documents. FiD-TF utilizes cross-attention to identify and eliminate unnecessary tokens, while FILCO requires training a contextual filtering model. These methods offer higher precision in content elimination but are computationally costlier than passage-level strategies. While these advancements have improved the performance of RAG systems, challenges remain. Most studies use relevance as the sole quality measure and employ single-text granularity elimination strategies. There is a need to consider more domain-specific quality factors and implement elimination strategies at various text granularities in CPQA.

2.2 Generation Customization

LLM generator selection in CPQA needs to consider many factors to ensure RAG system performance. However, most studies focus on the accuracy of responses, ignoring users' other needs (e.g., users' data privacy). Commercial models like GPT-4 match human expertise in some Medical QA (Table 2.1), outperforming open-source models such as Llama 2 and Mistral [43]. Although commercial models seem to be the gold standard, smaller, open-source LLMs do not need to be accessed through third-party APIs, offering data protection and lower cost, which would be the preferred solution in domains with strict privacy requirements and economic constraints like healthcare. So, there is an urgent need to explore smaller open-source LLM alternatives that, when used in RAG, can achieve response accuracy comparable to that of commercial models.

Model	Date	USMLE	MedMCQA	PubMedQA
GPT-4	2023	86.1	73.7	81.2
Human (passing score)	-	60.0	50.0	-
Human (expert score)	-	87.0	90.0	78.0

TABLE 2.1: Answering accuracy of LLMs VS Human [6]

Response adjustment involves negative rejection and enhancing the verifiability and comprehensibility of responses to meet the needs of CPQA users (patient customers). The negative rejection is that LLM rejects answering if the retrieved content cannot sufficiently answer the query [44]. Due to the potential life-critical nature of CPQA

queries [76], negative rejection is preferable in medical environments, as it minimizes the risk of providing potentially harmful information. Verifiability allows users to trace the sources of information and verify claims independently, which is also crucial because medical information will influence patients in critical decisions, diagnosis, and treatment [77, 78]. Comprehensibility is vital because the knowledge gap between medical information texts and patients impedes patient understanding and self-care [45, 79]. Specifically, consumers may misinterpret certain medical terminology, particularly for rare diseases, necessitating translating or simplifying the answer to enhance the comprehensibility of outputs [80]. Moreover, ensuring that the final response resembles human-generated text in terms of fluency is essential for improving comprehensibility. Currently, most machine-generated texts tend to be incoherent and produce insufficient evidence-based claims [81]. However, most current RAG-based medical QA systems are designed from a clinical perspective [39, 40], often neglecting patient consumers’ needs. So, this study investigates how to implement negative rejection and improve response verifiability and comprehensibility.

2.3 Non-single-step Retrieval

Active retrieval improving RAG system efficiency and response quality by correctly determining when retrieval is needed [82]. RAG does not always enhance responses; it sometimes negatively impacts the original outcome, particularly when LLM’s inherent parametric knowledge suffices for the query [83]. This occurs because LLMs tend to rely on provided retrieval results, impairing creativity and versatility, even accuracy [55]. Many active retrieval approaches have been proposed; for example, FLARE triggers retrieval when the confidence of generated terms falls below a threshold to regenerate sentences containing low-confidence tokens [82]. Another research [56] uses statistical question analysis to decide retrieval timing because it found that LLMs alone can effectively answer questions containing high-popularity entities. However, these statistical analysis-based methods often exhibit high domain specificity (e.g., the thresholds and definitions of high-popularity entities vary in different domains), presenting scalability challenges. A more popular research direction is determining retrieval methods by LM. For example, AdaptiveRAG, which trained an LM to enhance the RAG systems complexity-awareness(ability to discern query complexity), classifies retrieval actions based on query complexity [31]. However, it introduces additional LM predictors and ignores that some low-complexity queries may also need retrieval when they involve rare research like “Tuberous Sclerosis” [84] for which the LLM’s pre-training data may lack relevant knowledge. SKR leverages the LLM’s self-awareness (ability to assess whether it can answer a query) to bypass retrieval when confident, outperforming CoT-based and

fully retrieval-based methods with advanced models like InstructGPT or ChatGPT [83]. Self-RAG allows LLM to introspect its output based on “reflection tokens,” which involves using LLM’s knowledge-awareness (ability to assess whether supportive knowledge is needed) to trigger retrieval when necessary [55]. However, it relies on reflection tokens’ quality, requiring expensive LLM training. Overall, these active retrieval methods focus either on complexity-awareness or self-awareness or knowledge-awareness. Besides, most of them only consider when to retrieve, lacking the ability to process complex queries requiring multi-step retrieval.

Multi-step retrieval iteratively searches new queries generated according to search results from the previous step [85]. The two primary strategies are Iterative Retrieval and Recursive Retrieval. Iterative Retrieval integrates generated text into original queries, strengthening subsequent responses by providing LLM with expanded knowledge but potentially causing context overload [86]. Recursive Retrieval hierarchically explores nodes related to previous results, gradually converging on relevant information while maintaining context relevance but requires a hierarchical structure [85]. For example, in COMP90005, HSRT’s semantic search applies recursive retrieval: the MedCPT [87] first narrows candidates based on abstract-query similarity, followed by the passage retrieval within selected documents. Prompting techniques like Chain-of-Thought (CoT) improve response quality by guiding step-by-step reasoning [1, 88]. However, single-step retrieval based solely on the given query often captures only a fragment of the necessary reasoning for complex queries. So many methods integrate multi-step retrieval with prompting and reasoning techniques. For example, IRCOT interleaves retrieval with CoT reasoning, iteratively refining queries [58]. ITERRETGEN also iterates collaborative retrieval and generation but treats retrieved knowledge holistically to reduce overheads [89]. The Tree of Clarifications (TOC) method disambiguates unclear queries through recursive retrieval, refining them into specific sub-queries for processing [90], which is useful in fields like biomedicine, where users (e.g., patients) may have unclear information needs. Multi-hop retrieval explains evidence at each step and identifies bridge entities for broader exploration, enabling broader exploration compared to Recursive Retrieval’s reliance on direct associations [91]. These methods, however, increase latency due to multiple retrieval steps. Moreover, irrelevant retrieval results in any step can lead to cascading errors [59].

Therefore, it is critical to explore novel multi-criteria adaptive RAG systems to determine what type of retrieval (no retrieval, single-step retrieval, or multiple-step retrieval). This system’s criteria selection should consider CPQA’s needs, and the multiple-step retrieval should minimize latency and cascading errors.

Chapter 3

Data Collection and Analysis

3.1 Data Collection

The retrieval data collection is described in section 3.2.1. We constructed cancer-related QA datasets using a MeSH-based filter method. Medical Subject Headings (MeSH) [92] is a controlled vocabulary used to index and search biomedical literature databases. This approach utilized terms under the “neoplasm” MeSH subtree to identify cancer-related content, including synonyms and specific medical terms like “Tuberous Sclerosis.” The multiple-choice questions can simplify evaluation, eliminate biases from text similarity computations or human annotation, and align with large-scale medical QA system assessments [93–97]. Therefore, we use the following QA datasets (Figure 3.1) and put the introduction of original QA datasets in Appendix:

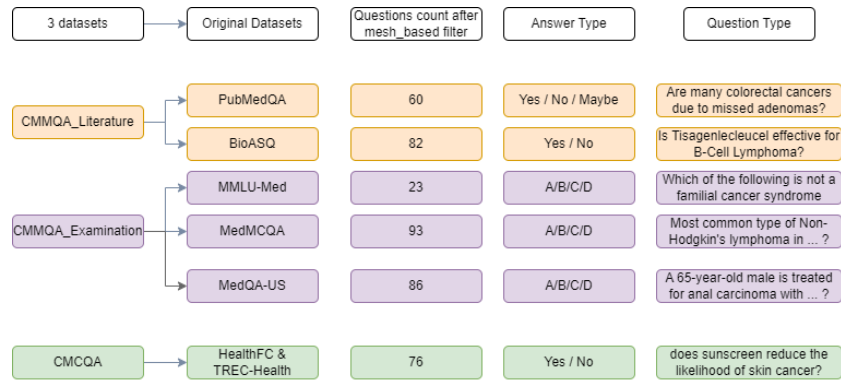


FIGURE 3.1: Introduction of CMMQA and CMCQA

Cancer-related Multiple-choice Medical QA Dataset (CMMQA) explores CPQA performance on complex questions from patients with prior medical knowledge. Besides, it is COMP90005’s evaluation set, enabling a direct comparison of CPQA accuracy.

Cancer-related questions were selected using the MeSH filter from five common medical QA datasets (MMLU-Med [98], MedQA [99], MedMCQA [100], PubMedQA [101], BioASQ [43]).

Cancer-related Multiple-choice Consumer QA Dataset (CMCQA) assess CPQA performance on common consumer health concerns in section 5.3.1. The dataset was constructed using the MeSH filter from two common customer QA datasets: HealthFC-2 [102] and TREC-Health [80].

3.2 Analysis Method

To avoid the subjectivity of human evaluation and due to resource and time constraints, automatic quantitative evaluation is employed to assess the CPQA performance and efficiency, enabling reliable and efficient evaluation. For multiple-choice datasets like CMMQA and CMCQA, the correctness of choice indicates if the response is correct in terms of factual information and appropriate in the context of the query [103]. So, we use exact matching to calculate the generation accuracy. Accuracy is the primary evaluation metric for these multiple-choice datasets for several reasons. First, in multiple-choice questions with only one correct answer and where the model is allowed to select only one option, the dataset is inherently balanced, as each option has an equal theoretical chance of being selected. Second, although multiple options exist, the task can be effectively reduced to a binary classification problem (whether the correct option is chosen or not). In such balanced binary classification tasks, Accuracy and micro F1 score are mathematically equivalent. Finally, Accuracy provides a simple, intuitive metric (the proportion of correctly answered questions) that is easy to calculate and interpret.

However, different tasks may require alternative evaluation metrics to address specific concerns. For instance, recall is the primary focus in negative rejection, measuring the proportion of correctly identified negatives (those that should be rejected). Maximizing recall is crucial for medical consulting, where false negatives (incorrect answers) carry higher risks than false positives. This trade-off, though reducing precision (more unnecessary rejections), aligns with the principle of “primum non nocere” [104] (first, do no harm). Other metrics, like accuracy, may mislead due to class imbalance, and while F1 balances precision and recall, recall should take precedence in life-critical contexts.

Chapter 4

Single-Step Comprehensive Retrieval With Evidence-Processing

4.1 Introduction

Here is the framework of **Single-Step Comprehensive Retrieval With Evidence-Processing** (SCREP). In this chapter, we will first introduce how to get comprehensive retrieval results, then describe how to process evidence:

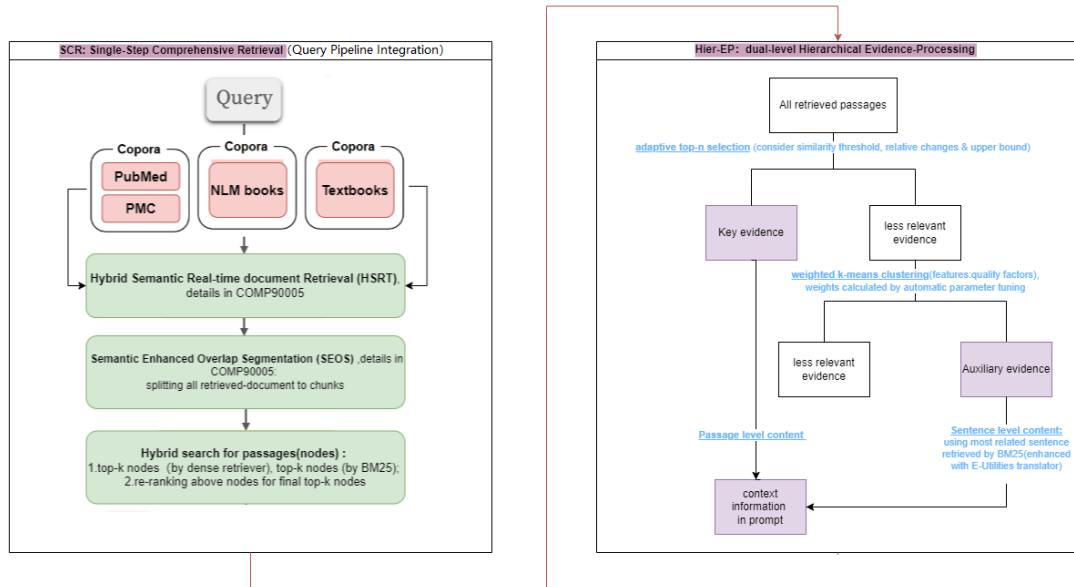


FIGURE 4.1: Framework of SCREP

4.2 Method

4.2.1 Query Pipeline integration

Expanding Sources Textbooks source [105] is a collection of 18 widely used medical textbooks for preparing USLME. We also introduced the NLM Books source, a publicly accessible subset of the total collection of books and documents on the NLM bookshelf. It contains 9330 public articles from Startpearls, a point-of-the-care clinical decision support tool similar to Up-To-Date. Unlike the process for online NCBI sources, which allows query-related document retrieval via E-Utilities, we downloaded new integrated resources. Given the length of the books and the fact that the cancer section is likely to appear in most books in different proportions, the books and documents must be segmented and filtered for cancer-related content. We first downloaded NLM Books and Textbooks, then utilized the existing hierarchical structure within the raw texts for chunking. Each article paragraph was treated as a segment, and all relevant hierarchical headings were concatenated to form corresponding titles. Subsequently, we employed the MeSH filter method to select cancer-related segments. Finally, we generated embeddings for these segments using the MedCPT Article Encoder and established an FAISS (Facebook AI Similarity Search) index [106], maintaining consistency with the document processing method for online NCBI sources in COMP90005. Hyperparameter selection is crucial in RAG and should depend on dataset analysis. We set the retrieval quantities ratio according to Table 4.1:

Source	FAISS Index	Avg. L	Domain
PubMed	34.2GB	296	Biomed, Online NCBI Source
PMC	-	-	Biomed, Online NCBI Source
NLM Books	0.48GB	118	Clinics, Medicine
Textbook	0.048GB	81	Clinics, Medicine

TABLE 4.1: Data Source Statistics, FAISS Index refer to the FAISS index size corresponding to the filtered text(PubMed abstracts in recent ten years, cancer-related segments from NLM books and Textbook)

The initial FAISS index ratio of Online NCBI sources: NLM books: Textbooks was 712.5:10:1 (Table 4.1), reflecting a significant imbalance in the raw data volumes. To mitigate this imbalance while preserving the relative significance of each source, we employed a logarithmic transformation with an offset ($\log(x) + 1$). This transformation resulted in an intermediate ratio of approximately 3.85:2:1. To further optimize the retrieval process and enhance the system’s performance, we adjusted this ratio to 4:2:1. This adjustment allows for more streamlined computation while ensuring the larger NCBI dataset contributes substantively without overshadowing other vital

sources. Subsequently, considering that only two-thirds of PubMed documents contain abstracts, which are crucial for meaningful retrieval, we further adjusted the ratio to 6:2:1. This final adjustment accounts for the effective information density in the NCBI dataset, ensuring that the retrieval process compensates for the absence of abstracts in a significant portion of the documents. This multi-step ratio adjustment process balances dataset influences and accounts for the nuanced characteristics of biomedical literature.

Single-Step Retrieval Strategies In hybrid search, fusion algorithms combine the results of different search models. Figure 4.2 briefly introduces two fusion algorithms. We observe that RRF solely focuses on the ranking position, neglecting the search relevance scores. Moreover, Relative Score Fusion requires extensive trials to optimize weight allocation. Therefore, we select reranker-based fusion to obtain the hybrid search results. So, the first group of experiments will use HSRT for Online NCBI sources as in COMP90005 while employing MedCPT-based document retrieval for NLM textbooks and Textbooks. After getting all the target documents, we applied SEOS, as mentioned in section 2.1.2, to split the retrieved documents into passages. Finally, we utilized the dense retriever (same as in COMP90055) and BM25 retriever to select the top k nodes for each retriever. We deduplicated the combined results and applied MedCPT reranker to select the final top k nodes. Using the source metadata of top k nodes, we can investigate the search preferences of different evaluation datasets towards various data sources in hybrid search, which can be used as a heuristic rule for decision-making in Route search. Since the cost of a hybrid search for the single-step search is lower than that of employing an LLM-based decision-maker in the Route method, we only consider integrating the Route method in the multi-step search.

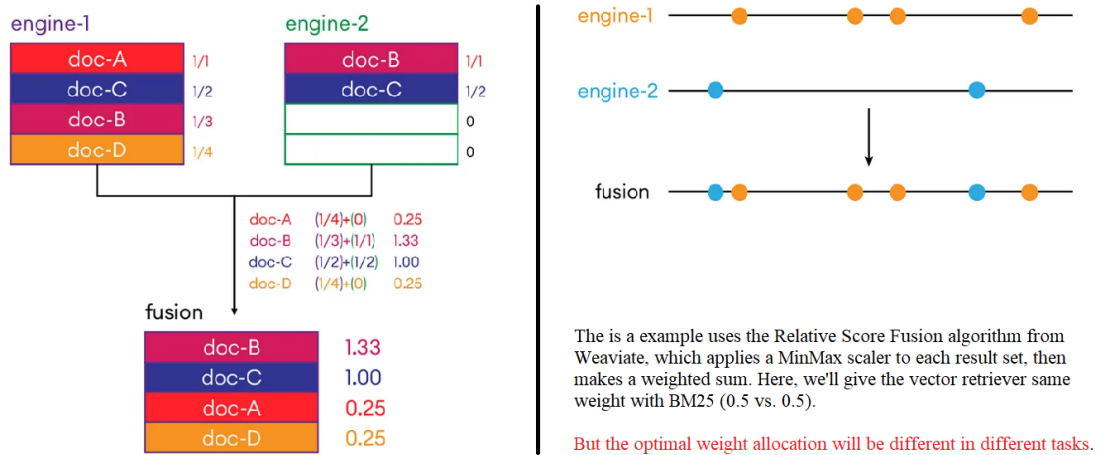


FIGURE 4.2: Left: Reciprocal Ranked Fusion (RRF) [3], Right: Relative Score Fusion

4.2.2 Hierarchical Evidence-Processing (Hier-EP)

Retrieved evidence quality is crucial for response accuracy. However, processing text at the token level is computationally expensive and can disrupt the meaning of sentences if necessary tokens are removed. So, we propose a novel dual-level **Hierarchical Evidence-Processing** (Hier-EP) algorithm that works at both paragraph and sentence levels.

Before evidence filtering, we use the ‘small to big’ principle when necessary. For example, Consider adding context for chunks only when the corresponding passage text does not exceed 32 tokens. This strategy preserves meaning while optimizing processing resources. Then we consider evidence-filtering, which has two aspects:

“Key evidence” - highly relevant information that significantly impacts response quality. Since query-related information occupies a large proportion of “key evidence” and may have critical semantic connections, all content of “key evidence” should be retained, thus not processed at the sentence level. Recognizing that fixed top-k selection methods lack flexibility, we investigate methods for dynamically selecting “key evidence.” Adaptive top-k selection with a self-reflection mechanism is costly because it requires submitting all search information to the LLM. So, we consider a data-driven adaptive top-k selection method. A simple baseline method uses statistical analysis to determine a similarity threshold. However, sole reliance on similarity threshold may miss specific scenarios (e.g., all evidence of a query exhibits relatively low similarity scores). Therefore, we suggest calculating the relative change in adjacent similarity scores and identifying significant differences. To maintain efficiency, we also use an upper bound that limits the number of “key evidence.” For instance, with a total evidence of 24 and an upper bound of 12, we select the top-n key evidence pieces, where n is determined by the dual consideration of similarity scores and relative changes, up to a maximum of 12.

“Auxiliary evidence” - high-quality but less relevant evidence that will undergo further processing at the sentence level, while other less relevant evidence will be dropped directly. Our parameter analysis encompasses multiple factors: 1) Relevance and evidence Length: These metrics can be calculable during retrieval. In COMP90005, we observed that longer texts contain more information and typically show lower relevance. 2) Publication Year and Citations: We can access the publication year using PubMed’s metadata. However, citation counts require additional calls to the Semantic Scholar API. Recently published papers provide the latest opinions, while higher citation counts indicate recognition. However, there is an inverse relationship between publication time and citation count—older books have more time to accumulate citations. 3) Data source: We differentiate between high-density knowledge sources (e.g., NLM books and textbooks) and more diverse (in quality) online repositories (e.g., NCBI sources). This distinction

informs our evidence classification process. Less relevant evidence from high-density knowledge sources will add to the “auxiliary evidence” set, but evidence from online NCBI sources will be classified while considering optimal factor combinations. This is an unsupervised classification task; we use k-means clustering [107] to predict whether an evidence passage is “auxiliary evidence.” In k-means clustering, we use factor combinations to adjust the importance of each feature. Specifically, each feature (cosine similarity, text length, publication year, and citation count) is multiplied by weight, helping the clustering process focus more on the important features. The weighted features are then scaled and used to fit the k-means model, which groups the data into clusters. To explore optimal factor combinations, we consider three methods: a) Grid search: This method, though straightforward, requires extensive experiments to determine the combination of four factors and may still yield inaccurate results. b) Mathematical modeling: This method is challenging because the factors interact in complex ways. c) Automatic parameter tuning inspired by DSPY [108] It selects accuracy as the optimization target and employs the Adam optimizer, which has dynamic parameter adaptation capabilities and implements warm restarts where the learning rate periodically increases and then decays [109], to adjust four parameters to achieve the highest accuracy. This method is similar to how attention mechanisms work in neural networks, where attention weights are learned automatically to focus on the input’s important parts [110].

Sentence Level Processing: Previous studies [111, 112] used sparse retrievers for lexical information at the sentence level while dense retrievers at the passage level; this is because BM25 is a proven method for efficient and effective sentence retrieval method, while dense retrievers require encoding text information, making dense search for sentences uneconomical. Thus, we also use BM25 as a sentence retriever to find the most related sentence per “auxiliary evidence,” enhancing its capture of semantic meaning with E-Utilities’ built-in translator for query optimization (Figure 4.3).

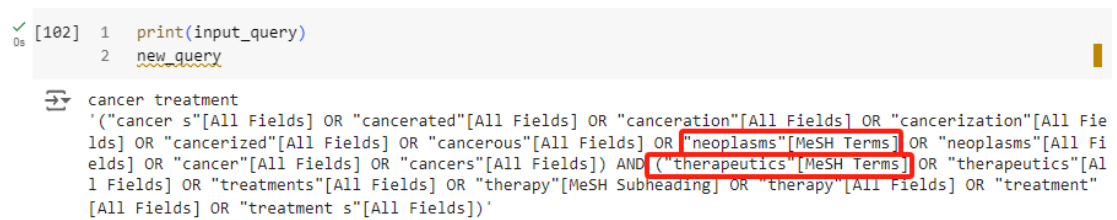


FIGURE 4.3: Utilizing E-Utilities’ built-in translator for mapping original terms to their corresponding MeSH terms, synonyms, and lexical variants

The selected sentences and the entire “key evidence” text are passed to the LLM.

4.3 Experimental Results and Analysis

We employed the concept of hard negatives [113], constructing a hard negative test dataset consist CMMQA questions incorrectly answered by both RAG in COMP90005 and baseline method (LLM+COT). Due to the limited number of MMLU-Med negative questions (only 3), we excluded them from our analysis. To compare the results of the best query pipeline in COMP90005, we use the same generator - Claude-3-Haiku.

4.3.1 Query Pipeline integration

Method	Negative Cancer QA Dataset			
	PubMedQA	MedMCQA	MedQA	BioASQ
Expanding Sources(ratio set 1)	11.11	28.00	28.00	25.00
Expanding Sources(ratio set 2)	16.67	28.00	24.00	33.33
+ Hybrid Search	16.67	36.00	32.00	41.67

TABLE 4.2: Experiment Results(ACC); Online NCBI sources: NLM books: Textbook, ratio set 1 is 1:1:1, ratio set 2 is 6:2:1 (discussed in section 4.2.1), the total count of the retrieved document is 72

In the default experiment condition, Table 4.2 and Figure 4.4 underscore the importance of diverse, comprehensive knowledge sources in medical QA systems. Table 4.2 showcases Expanding Sources corrects erroneous responses by leveraging additional knowledge, demonstrating improved performance with larger corpora. Hybrid passage search further enhances retrieval by capturing content dense retrievers might miss. Our experiments reveal that comprehensive single-step retrieval improves performance by surfacing more relevant passages. Figure 4.4 shows that Online NCBI sources, including PubMed and PMC (two large corpora), prove suitable for various medical questions. While we do not recommend using single corpora outside of online NCBI sources for medical QA due to limited medical knowledge and inability to access the latest content or research, they may prove beneficial for specific tasks (e.g., QA in medical examination sets containing more clinical questions). This aligns with previous research showing improved performance on clinical questions when retrieving from medical textbooks [61]. This finding may explain why using ratio set 2 results in an accuracy increase on PubMedQA and BioASQ but a decrease in accuracy on MedQA, compared to using ratio set 1. Another interesting finding is that NLM books and Textbooks, despite their smaller index size (1/700 of online NCBI source), contribute a higher proportion of top k passages than expected, suggesting that NLM books and Textbooks may possess higher information density compared to documents from online NCBI sources.

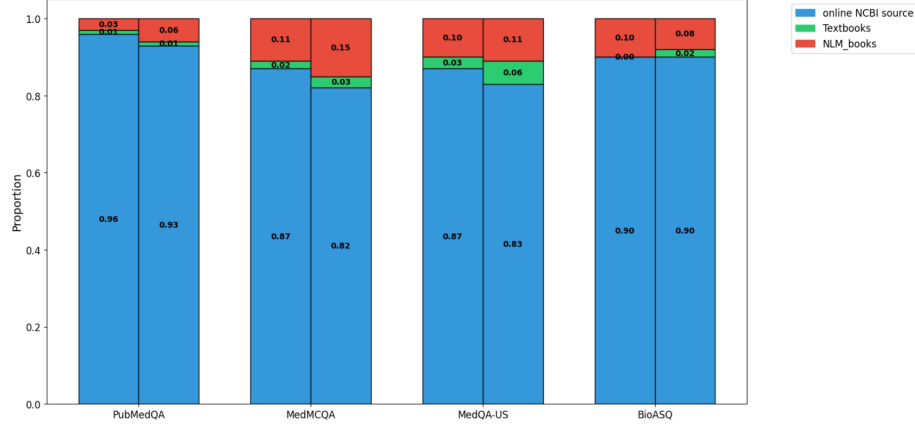


FIGURE 4.4: Ratio of retrieved evidence from three biomedical corpora using the best method in this section. In subfigure, left: in top-6 evidence, right: in top-12 evidence.

4.3.2 Hierarchical Evidence-Processing(Hier-EP)

We began by employing an automatic parameter-tuning process to determine the optimal factor combination for our algorithm. This process involved iterative testing and validation to ensure robustness. The optimal parameters were then applied in a k-means clustering algorithm to identify “Auxiliary evidence.”

Method	Negative Cancer QA Dataset			
	PubMedQA	MedMCQA	MedQA	BioASQ
fixed top-n method	16.67	36.00	32.00	41.67
Only “key evidence”	11.11	36.00	36.00	50.00
Hier-EP algorithm	22.22	40.00	36.00	50.00

TABLE 4.3: Performance(ACC) of different methods, experiment default parameters setting: within top-12 evidence, similarity threshold = 0.5 , compression rate = 1

The Hier-EP algorithm outperforms others (Table 4.3) because it uses both “auxiliary evidence” and “key evidence.” The “auxiliary evidence” component ensures that critical details in newer studies or highly cited papers, which may provide unique insights (Figure 4.4), are not overlooked. Interestingly, using only “key evidence” can sometimes even improve performance compared to fixed top-n methods. This might be because focusing solely on “key evidence” alleviates the LLM’s “distraction phenomenon,” although it may ignore potentially helpful details in “auxiliary evidence.” The nature of the QA dataset also plays a role. Medical examination datasets’ questions focus on classic knowledge rather than the latest research findings, so ignoring “auxiliary evidence” may only cause limited influence. Although both PubMedQA and BioASQ are biomedical research QA datasets, the questions in PubMedQA are manually created based on existing research article titles, and the long answers are the conclusions of the abstracts. So, in PubMedQA, the answer to certain questions may come from a specific relevant

sentence in the snippet, emphasizing sentence-level relevance. In contrast, BioASQ questions are designed for biomedical experts’ real information needs, making passage-level relevance more critical, requiring more attention to “key evidence.”

Question	Document #1	Document #2
Can stress promote dementia? (Supported)	To test the hypothesis that high job stress during working life might lead to an increased risk of dementia and Alzheimer’s disease (AD) in late life. (...) Lifelong work-related psychosocial stress, characterized by low job control and high job strain, was associated with increased risk of dementia and AD in late life, independent of other known risk factors. Wang et al. (2020) (Supported) [121 citations]	Patients with Alzheimer’s disease (AD) or dementia are increasing in numbers as the population worldwide ages. Mid-life psychological stress, psychosocial stress and post-traumatic stress disorder have been shown to cause cognitive dysfunction. The mechanisms behind stress-induced AD or dementia are not known. Zhu et al. (2021a) (Refuted) [3 citations]

FIGURE 4.5: Newer or highly cited studies’ positive impact on QA performance [4]

For the Hier-EP(upper-bound, total count) functions, where total count represents the total evidence count and upper-bound ensures that “key evidence” does not exceed the upper-bound, the algorithm’s performance (Figure 4.6) can be analyzed as follows: For a fixed total count, using a lower compression ratio reduces the tokens that the LLM generator processes. This reduction in token count correlates with shorter generation times for answers on the Negative Cancer QA Dataset. However, when evidence is overly compressed, the granularity of texts passed from lower-ranked “key evidence” to prompt transitions from passage-level to sentence-level, or lower-ranked key evidence may be entirely discarded, contributing to a decline in performance. The average response accuracy peaks when the upper-bound is approximately 12, suggesting a potential optimal point that balances efficiency and response accuracy.

Since there are two accuracy peaks, we provide details in Table 4.4 to explore the impact of different parameter settings on the performance of the Hier-EP method across various datasets. When increasing the total count to 24 while maintaining the upper bound at 12 (comparing Hier-EP(12, 12) and Hier-EP(12, 24)), the LLM generator processes more “auxiliary evidence,” resulting in improved performance on PubMedQA and MedMCQA but decrease accuracy on BioASQ. Meanwhile, decreasing the upper bound from 12 to 6 (comparing Hier-EP(12, 12) and Hier-EP(6, 12)) allows the LLM generator to focus more on the most relevant information, leading to improved performance on MedMCQA, MedQA, and BioASQ. This demonstrates that the nature of the QA dataset influences the importance placed on “key evidence” and “auxiliary evidence.” PubMedQA, requiring more sentence-level information, benefits from greater emphasis on “auxiliary evidence.” Meanwhile, MedQA and BioASQ prefer the LLM to focus on

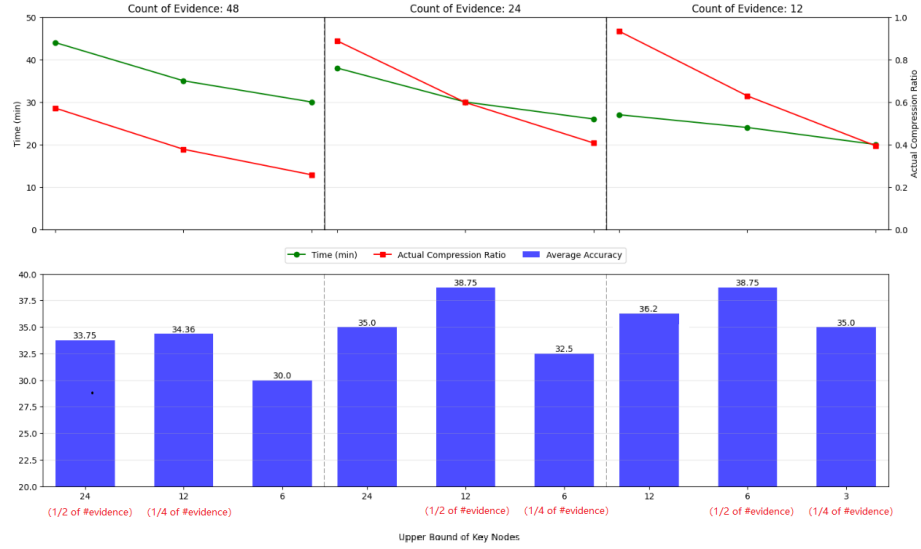


FIGURE 4.6: Parameter's impact on efficiency and accuracy

the most relevant paragraph-level information. This further illustrates that the optimal parameter configuration is dataset-specific, highlighting the importance of tuning parameters based on the characteristics of the specific dataset or query.

Method	Negative Cancer QA Dataset				Avg
	PubMedQA	MedMCQA	MedQA	BioASQ	
Hier-EP(12, 12)	22.22	40.00	36.00	50.00	36.2
Hier-EP(12, 24)	27.78	48.00	36.00	41.67	38.75
Hier-EP(6, 12)	22.22	44.00	40.00	50.00	38.75

TABLE 4.4: Details of response accuracy in different datasets when changing parameters of Hier-EP(upper-bound, total count) function

Chapter 5

Generation Customization

5.1 Introduction

For patients’ data privacy, we will employ a rigorous evaluation framework to compare the performance and efficiency of various LLMs using different methods on diverse datasets, thereby identifying the most suitable LLM generator for CPQA. Subsequently, we will adjust response to cater to patients’ additional needs in negative rejection, comprehensibility and verifiability of response.

5.2 Method

5.2.1 LLM Generator Selection in CPQA

We selected several prominent LLMs in recent years (Table 5.1). Initially, we conducted baseline experiments using only CoT prompting. The response accuracy of these models on the CMMQA dataset will be used to select LLM generator candidates in CPQA.

Model	Context	Size	Release	Open	Domain
PMC-LLAMA	2K	13B	2023	Y	Biomed.
Llama-2	2K	13B	2023	Y	General
Mixtral	32K	46.7B	2023	Y	General
Llama-3	8K	70B	2024	Y	General
Llama-3	8K	8B	2024	Y	General
Claude-3-Haiku	200K	20B	2024	N	General
GPT4	32K	1.76T	2023	N	General

TABLE 5.1: Prominent LLMs and their characteristics

Then, we assessed performance improvements using SCREP with different generators. We recorded evaluation times, initial confidence, answer choices, and final confidence

for baseline and SCREP RAG configurations across different LLMs (Figure 5.1). Our analysis focused on the impact of SCREP RAG on response accuracy, evaluation time, self-awareness accuracy, and reflection accuracy (posterior judgment accuracy).

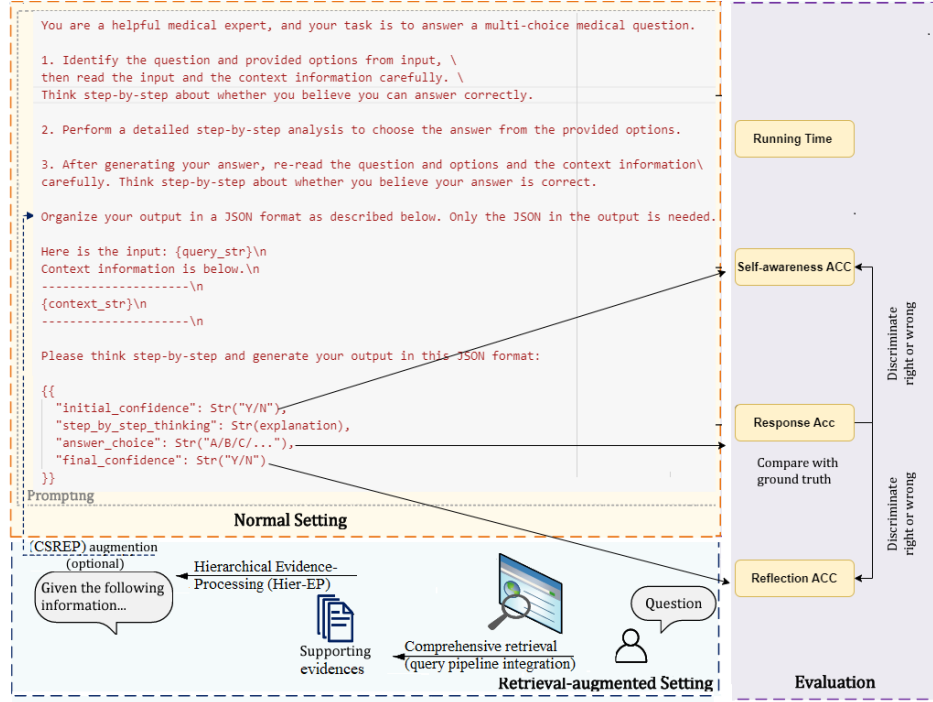


FIGURE 5.1: Evaluation process

5.2.2 Response Adjustment

We will analyze two methods to implement negative rejection: 1) utilizing the LLM’s self-reflection and 2) utilizing the LLM’s prior judgment. Then, we will consider a more helpful method for negative rejection to handle all kinds of queries in section 6.2. The comprehensibility and verifiability of responses are also critical aspects of generation customization. Our baseline utilizes the “step-by-step thinking” section in the prompt in Figure 5.1, primarily designed for clinical QA without specific optimizations for patient consumers. We enhanced RAG with Comprehensibility and Verifiability Boosters (CVB) using the following methods: **1) Source Attribution:** The model is instructed to highlight sentences generated from its own knowledge in red and search the prompt for related context information to regenerate the sentence. Sentences generated from the prompt context information are highlighted in green, and the corresponding reference number is added. **2) Citations:** For user convenience in verification, we adopt a particular format: reference number, URL, and evidence text. Users can click the URL to view the full text of the reference for more information. Moreover, they can only read the provided evidence text to validate whether the evidence text supports sentences using its reference number. **3) Readability Adjustment and Fluency Optimization:**

we add a ‘patient-friendly-explanation’ part which is simplified ‘step-by-step thinking’ explanation. This user-friendly addition improves comprehensibility through clear and specific guidance (e.g., everyday language style, short sentences, given structure) in the prompt. Considering that the average readability level of a US adult is at the 7th-8th grade level [114], we employ the Flesch-Kincaid Grade Level formula [115] to iteratively refine the ‘patient-friendly-explanation’ until the desired readability is achieved. Additionally, we require the generated text to be fluent and natural. **4) Medical Terminology Simplification:** We use LLM to identify complex medical terms and utilize columns (Preferred Label, Synonyms, Definitions) in the “neoplasm” MeSH subtree to provide complex medical terms’ corresponding definitions. Then, we encourage the LLM to provide a simplified explanation of the medical terms and corresponding definitions, ensuring that most complex medical terms can be understood.

5.3 Experimental Results and Analysis

5.3.1 LLM Generator Selection in CPQA

Using both Table 5.2 and Table 5.1, we can find some relations between performance and characteristics of LLMs: **1) Parameter size:** A straightforward idea is that larger parameter sizes can offer higher performance, demonstrated by the outstanding performance of the 1.76T GPT4 and Llama-3-70b. However, parameter size is not the only determinant of performance. For instance, Llama-3-8b, with fewer parameters than Llama-2-13b, outperforms the latter across all datasets. This suggests that other factors, like model architecture and training data, also play crucial roles. **2) Domain-specific training:** PMC-Llama, based on Llama-2-13b and fine-tuned in the biomedical research literature, outperforms Llama-2-13b across all datasets and shows exceptional performance on PubMedQA. This highlights the importance of domain-specific fine-tuning, which leverages rich background knowledge to enhance model understanding and performance in specialized areas [94]. **3) Open-source LLMs VS Commercial LLMs** Commercial models tend to outperform open-source models. GPT4 shows the best performance but is significantly more expensive than other LLMs. Among commercial models, Claude-3-Haiku is a cost-effective choice. Its 200K context window size allows it to receive more retrieval information as a generator in RAG, which is why we have consistently used Claude-3-Haiku as the default generator in COMP90005 and previous experiments. However, we also observe that with the rapid development of open-source models, some (like Llama-3-70b) achieve comparable performance to commercial models. For CPQA (patient privacy is critical), smaller, open-source models that

run locally and offline are more favorable. We chose the latest generation open-source LLMs (Llama-3-8b and Llama-3-70b) as generator candidates in CPQA.

LLM+COT	CMMQA					Avg
	MMLU	MedQA	MedMCQA	PubMedQA	BioASQ	
PMC-LLAMA	52.17	45.35	45.16	53.33	53.66	49.13
Llama-2-13b	34.78	20.93	18.28	33.33	37.80	27.32
mixtral-8x7b	73.91	69.77	60.22	31.67	60.98	58.72
Llama-3-70b	<u>91.30</u>	79.07	79.57	<u>51.67</u>	<u>76.83</u>	<u>74.71</u>
Llama-3-8b	60.87	65.12	65.59	<u>51.67</u>	67.07	63.08
Claude-3-Haiku	78.26	68.60	65.59	45.00	80.49	67.15
GPT4	95.65	<u>75.58</u>	79.57	48.33	84.19	75.29

TABLE 5.2: Generation accuracy of LLMs+COT on the Filtered Cancer QA Dataset.

SCREP improves CPQA (generator: Claude-3-haiku) performance (Table 5.3), and this performance gain varied but consistently exists across most LLMs and datasets (Figure 5.2), with minor exceptions observed in Llama-3-8b and GPT-4 on the MedMCQA dataset, Llama-3-70b in CMCQA dataset. This finding highlights SCREP’s generalization ability and the need to customize it for datasets (like adjusting parameters of Hier-EP functions in section 4.3.2) and LLMs. Notably, SCREP RAG elevated Llama-3-8b’s average response accuracy above the original Claude-3-haiku, and elevated Claude-3-haiku’s average response accuracy above the original GPT-4, demonstrating that SCREP RAG can make smaller LLMs viable contenders for CPQA generators. The response accuracy improvement is because SCREP uses Hier-EP algorithm to select high quality evidence within comprehensive retrieval results, and subsequently, returns all content for “key evidence” while providing the most related sentence per “auxiliary evidence.”

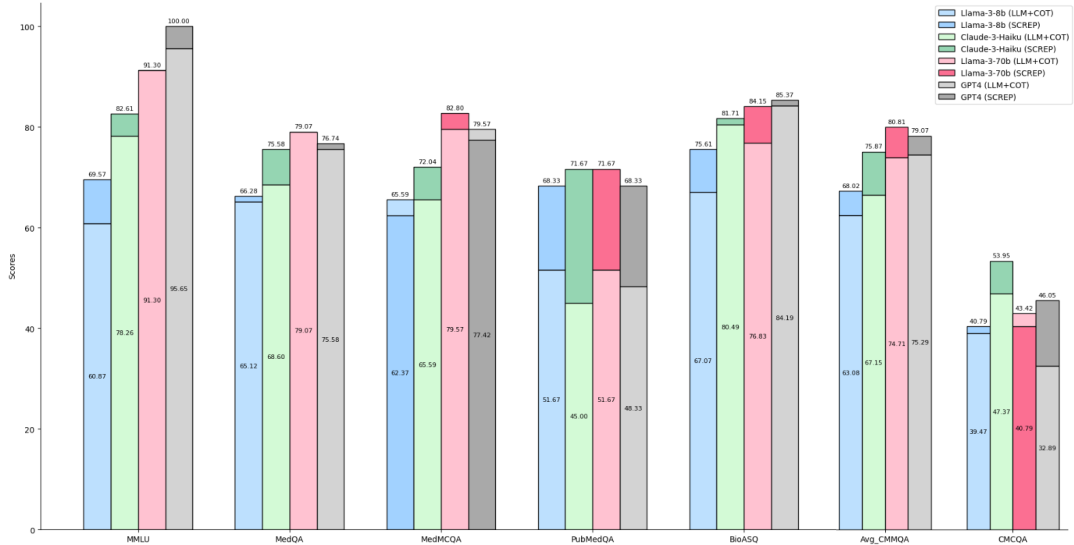


FIGURE 5.2: Response Accuracy Comparison

Furthermore, SCREP RAG improved not only response accuracy but also the accuracy of LLMs’ self-awareness (prior judgment) and reflection (posterior judgment), and the most

Method	CMMQA					Avg
	MMLU	MedQA	MedMCQA	PubMedQA	BioASQ	
LLM+COT	78.26	68.60	65.59	45.00	80.49	67.15
Standard RAG	82.61	67.44	65.59	56.67	81.71	69.48
RAG-COMP90005	82.61	69.77	67.74	66.67	81.71	72.39
SCREP RAG	82.61	75.58	72.04	71.67	81.71	75.87

TABLE 5.3: Comparison performance, Standard RAG: document retrieved pass to LLM directly, RAG-COMP90005: using best query pipeline in COMP90005

significant improvements in the smaller models like Llama3-8b (Figure 5.3). However, it substantially increased LLM’s processing time due to additional context information in the prompt. For instance, Llama3-70b’s average processing time per query increased from 4 seconds to 35.2 seconds when using SCREP RAG. So, using both Llama3-70b and SCREP in CPQA may decrease user satisfaction.

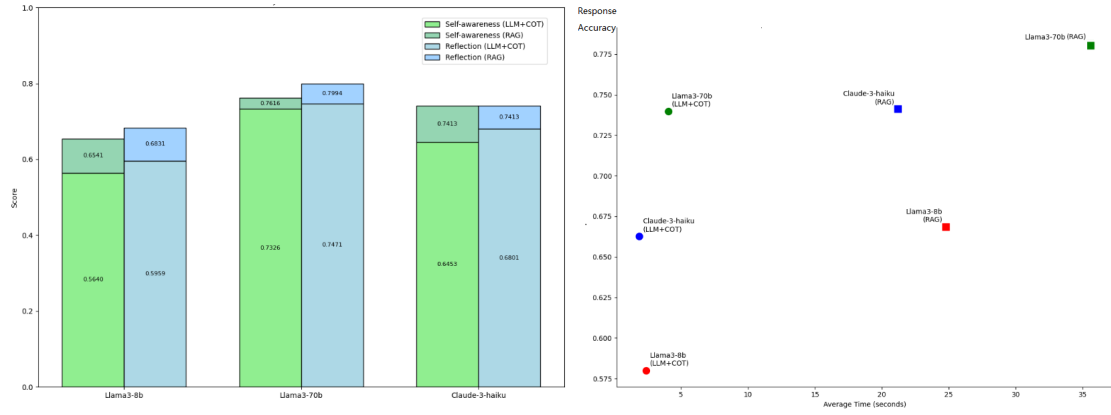


FIGURE 5.3: Impacts on multiple performance factors and generated time of SCREP

5.3.2 Response Adjustment

Acknowledging uncertainty can be as crucial as providing accurate information in medical contexts. Therefore, improving LLMs’ self-assessment is essential. Notably, self-awareness accuracy improvement is because RAG enhances LLMs’ understanding of factual knowledge boundaries [116], but reflection accuracy improvement is likely because RAG improves response accuracy and LLMs always have high post-task confidence levels (Table 5.4). A high post-task confidence level (rarity of rejections from posterior judgment) means a low recall of negative rejection when using the LLM’s self-reflection. Moreover, RAG enhances LLMs’ pre-task confidence (total rejections of prior judgment decrease), which may lead to a drop in recall of negative rejection using LLM’s self-awareness (prior judgment). This decrease in recall of negative rejection implies that more misleading or harmful information may be provided, making it unadvisable to rely on LLM’s prior judgment for negative rejection. Another interesting finding is the

discrepancy between LLMs’ confidence and actual performance. For instance, Claude-3-haiku exhibits the highest pre-task confidence levels but has a lower response accuracy than LLaMA-3-70B. Additionally, the noted bias in Prior judgment suggests that LLMs can suffer from similar cognitive distortions like the Dunning-Kruger effect characterized by underestimating abilities in some cases and overestimating in others[117, 118], underscoring the need to improve LLMs’ self-assessment further.

Model	Method	Prior-J		Post-J		SR	Recall
		CR	TR	CR	TR		
Llama3-8b	LLM+COT	37	87	2	2	127	0.291
	RAG	18	44	1	3	110	0.164
Llama3-70b	LLM+COT	21	47	0	0	87	0.241
	RAG	11	38	3	3	66	0.167
Claude-3-haiku	LLM+COT	4	15	0	2	113	0.035
	RAG	0	0	2	4	83	0.000

TABLE 5.4: Rejections by Prior Judgment (Prior-J) and Posterior Judgment (Post-J). CR: Correct Rejections, TR: Total Rejections, SR: supposed rejections. In this table, Recall: recall of negative rejection using prior judgment

Case study Comprehensibility and Verifiability Booster (CVB) ensures better communication between CPQA and patient consumers. We use the ‘step-by-step thinking’ and ‘citations’ part in Figure 5.4 for source attribution (green text: reasoning from external literature, , accompanied by reference numbers for verification; red text: reasoning from LLM’s internal knowledge) and provide a ‘patient-friendly-explanation’ part to help users quickly read and understand. Notably, even when the correct answer choice is identified, hallucination phenomena may still exist in the textual responses (red-boxed section of Figure 5.4 mentions significant toxicity, which is not supported by reference 2). These subtle differences in the text underscore the importance of source attribution and verification for AI-generated answers.

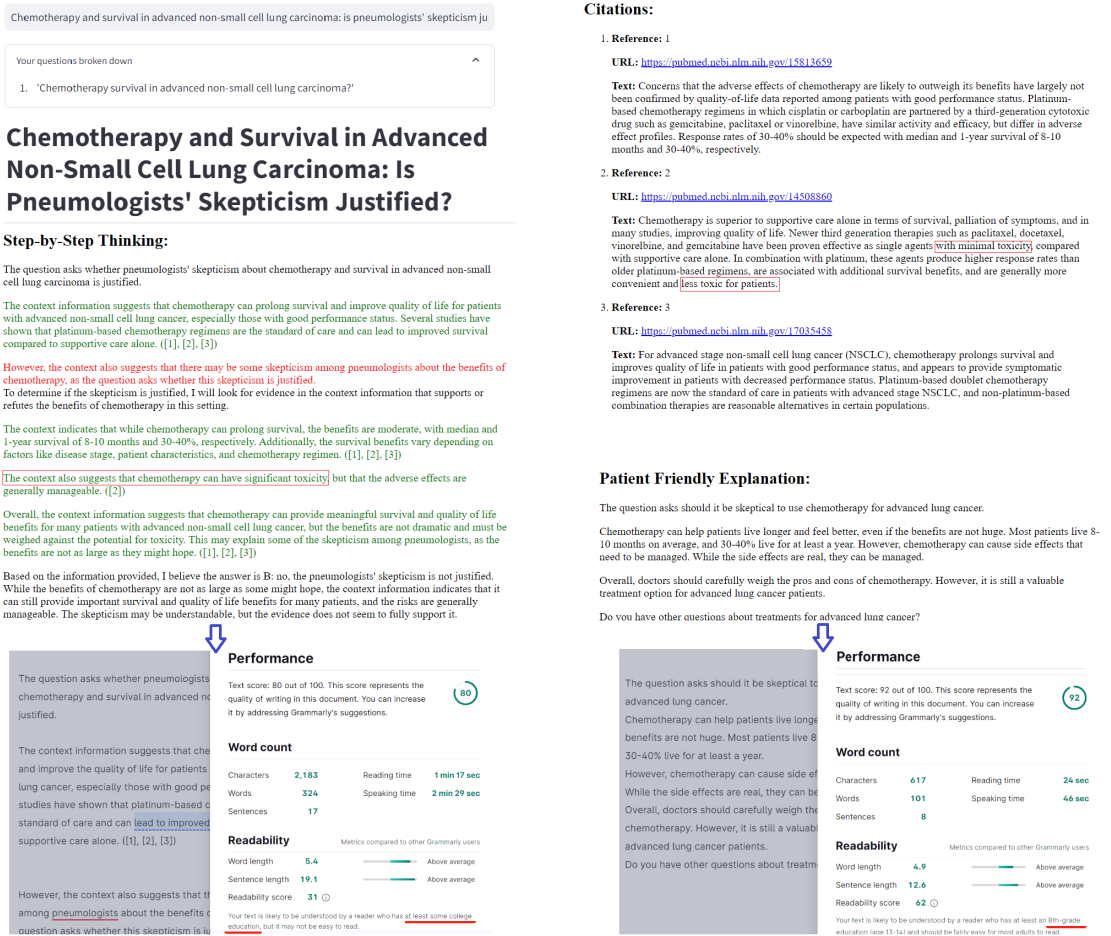


FIGURE 5.4: Case study of RAG with Comprehensibility and Verifiability Boosters (CVB), hallucination phenomena exists in the textual responses

Chapter 6

Multi-Criteria Adaptive RAG

6.1 Introduction

SCREP enhances CPQA but does not suit all user queries. For queries where LLM’s internal knowledge suffices, CSREP reduces efficiency and may introduce noise, lowering response quality. Hence, most active RAG systems emphasize retrieval timing. For complex queries, CSREP may capture only solution fragments, failing to provide correct answers and necessitating multi-step retrieval. To address these limitations, We propose a **Multi-Criteria Adaptive RAG (MCA-RAG)** to judge when and how to retrieve. In MCA-RAG, we introduced a novel **Dual-LLM Collaborative React-based Multi-Step Retrieval (DCRMR)** to combines the strengths of two LLMs: LLaMA-3-8B for generating per step, LLaMA-3-70B as the ReAct agent for three high-level reasoning and decision-making tasks (routing, query processing, and reflection) to reduce latency and improve retrieval quality. Finally, we explored how to implement negative rejection.

6.2 Method

MCA-RAG The criteria of MCA-RAG are: 1) Knowledge-awareness: Assessing whether the need for medical knowledge exists. 2) Self-awareness: Assessing whether the need for external knowledge exists. 3) Complexity-awareness: Determining if the question demands complex multi-step retrieval and reasoning (e.g., comparing multiple drugs). Other criteria, such as Intent-awareness (whether user intent for retrieval-augmentation exists) or Time-Sensitive-awareness (whether the need for real-time data exists) [119], are not considered to simplify the process. The exclusion of intent-awareness is due to the potential moderate to severe harm caused by incorrect answers to oncology queries [76]; even if the patient customer did not intend to use retrieval-augmentation, CPQA

should also consider retrieval-augmentation as an option. The exclusion of time-sensitive awareness is justified by the rapid evolution of medical knowledge and new clinical trial data need of oncology queries [120]. So, for all oncology queries, CPQA will use SCREP considering time as a quality factor or use multiple-step retrieval that uses the Hier-EP algorithm in every step. Following previous research [119], we trained two plug-and-play binary classifiers on the hidden states of the last layer of the LLM generator, allowing the classifier and generator to share the same input encoding (Figure 6.1). Compared to AdaptiveRAG’s classifier, which uses a different LM from the generator, our approach eliminates additional input encoding costs and avoids classification performance degradation due to inconsistent internal knowledge. To enhance the RAG system’s self-awareness and complexity-awareness, we followed previous work [31, 121] in training two classifiers using corresponding datasets. For the self-awareness classifier, we employ the LLM+COT method to get LLM-specific (positive label: I know) datasets, while we use the CSREP RAG method to get query-specific (positive label: can be answered by single-step approach) datasets for the complexity level classifier. We utilize 70% of the CMMQA dataset for training and designate the remainder as test data. During training, we freeze the LLM generator’s parameters and use it to capture complex semantic information in the text. We only update the parameters of the lightweight plug-and-play binary classifiers, avoiding LLM fine-tuning cost and catastrophic forgetting [122].

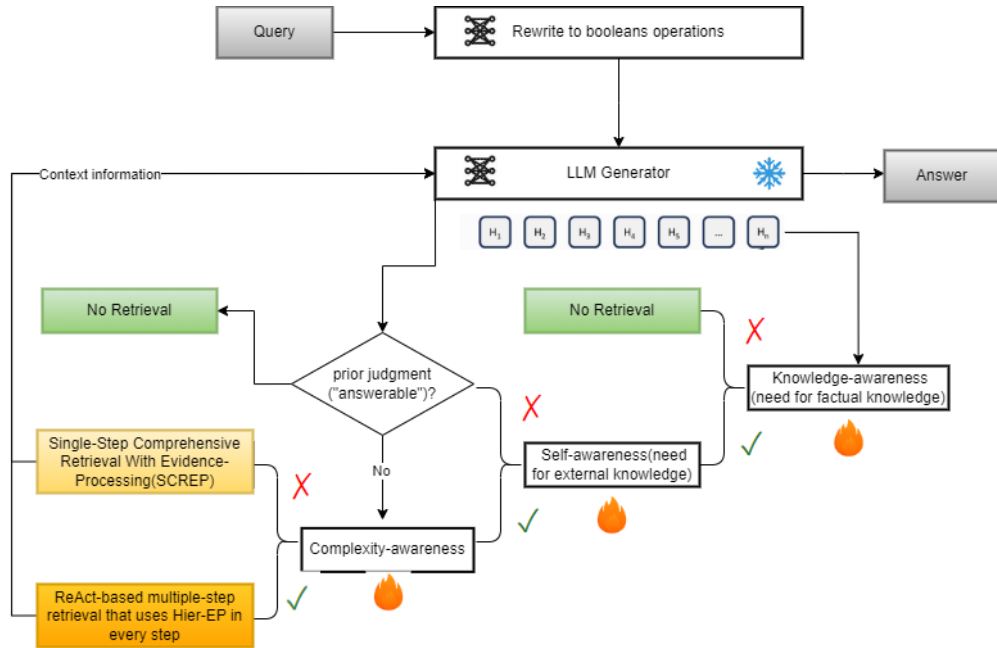


FIGURE 6.1: MCA-RAG integrates Knowledge-awareness, Self-awareness and Complexity-awareness: For knowledge-awareness, since CPQA provides patients with medical or emotional counseling, we use a few-shot prompt [5] to determine medical-related queries that need factual knowledge. Considering the importance of accuracy in oncology QA, retrieval is bypassed only when both the LLM prior judgment and the self-awareness classifier deem the given question answerable using the LLM’s internal knowledge. For queries classified as complex, we will use DCRMR.

DCRMR ReAct synthesis reasoning and acting, simulating the human thought-action-feedback loop to enhance LLM capabilities. In each iteration, the LLM agent combines cognitive reasoning and action planning, calls tool(s) to execute planned task-specific action, and uses action results for reliable observations. Compared to traditional multi-step retrieval methods in section 2.3, ReAct-based Multi-Step Retrieval is flexible because the ReAct agent can perform various actions to handle diverse tasks. However, this flexibility requires a more capable model as the ReAct agent. Section 5.3.1 shows that LLaMA-3-70B has a strong capability, but using it to generate answers in CPQA with SCREP will be slow. Compared to the original ReAct-based Multi-Step Retrieval using only an LLM (Figure 6.2), this study introduces a novel architecture: the faster, smaller LLaMA3-8B generates output in each retrieval step, while LLaMA-3-70B serves as the ReAct agent for high-level reasoning and decision-making:

1) Planning and Dynamic Routing: The agent analyzes the observation, identifies information gaps, and generates new sub-queries for additional information. The latter essentially is automatic query expansion, ensuring comprehensive information gathering. Finally the agent select appropriate tools for perform actions, we set three routing rules to optimize performance and minimizes latency. Firstly, retrieval source selection is based on the query type (clinical, biomedical, or others). Secondly, adapting the parameters of Hier-EP will consider specific information needs (sentence-level or term-specific evidence preference). Finally, after the current step has reached half of the total number of steps set, the self-awareness classifier and the LLM’s prior judgment are employed to determine when to bypass retrieval. This strategy is because defaulting to always external reliable sources or active retrieval will cause high latency or a high possibility of cascading errors.

2) Reflection Mechanism: The ReAct agent validates the consistency between step-by-step thinking and citations part, assessing whether all statements of answers can be supported by evidence (SUP). Based on SUP tokens and explanations, the agent decides whether to proceed to the next step or correct the step-by-step thinking part, mitigating cascading errors. When the ReAct agent determines that no further steps are needed, it evaluates whether all statements of answers are a useful response to the question (USE), deciding whether to use the answer directly, rewrite it, or reject it.

Negative Rejection Despite these advancements, some queries may remain unanswerable since using the three retrieval-augmented approaches may all fail to find related evidence (e.g., no related information in the source). We have analyzed 2 baseline methods in section 5.3.2; using the original LLM’s pre-task or post-task confidences can be unreliable. We recommend using the reflection mechanism for negative rejection. In DCRMR, if a sub-query exhausts all available tools and still fails to retrieve an answer, or if the final generated answer has the lowest USE score (indicating irrelevance to the

query), the system is designed to reject the answer. For queries that require only single-step retrieval, we apply a similar approach by introducing another LLM as “supervisor” to evaluate the SUP and USE score of the final answer and take appropriate actions based on this evaluation.

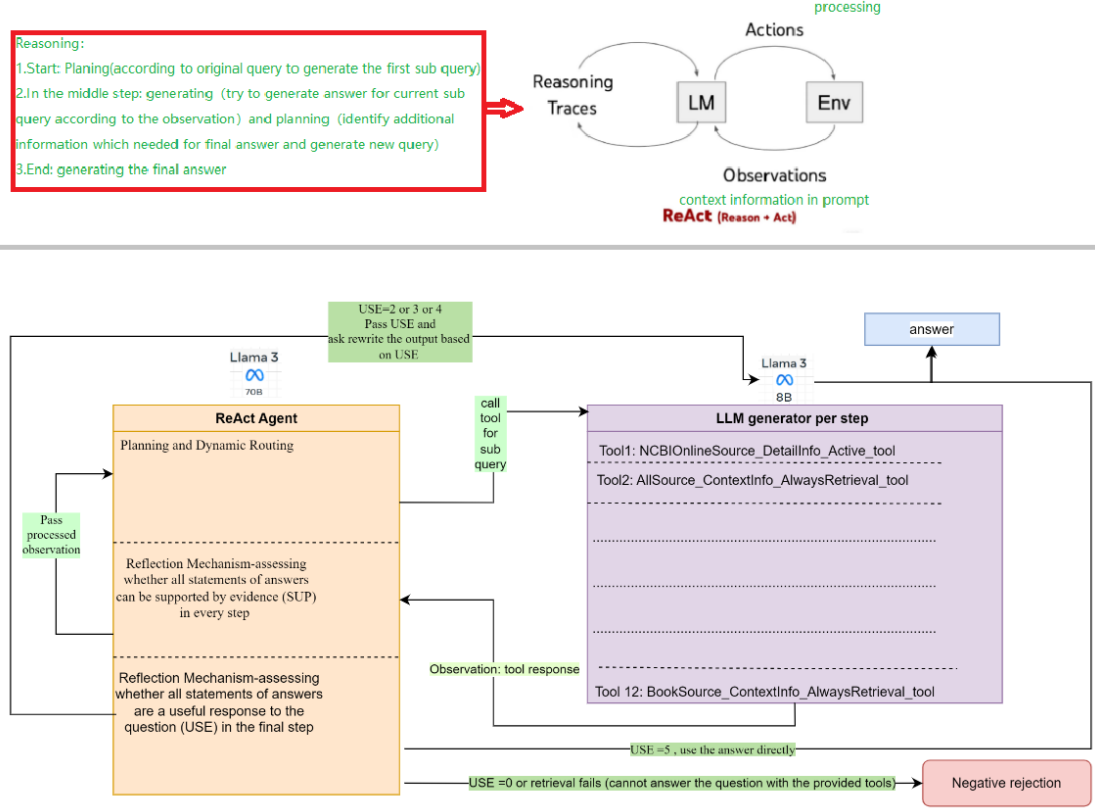


FIGURE 6.2: Upper: Original (Native) ReAct-based Multi-Step Retrieval. Down: Dual-LLM Collaborative ReAct-based Multi-Step Retrieval (DCRMR)

6.3 Experimental Results and Analysis

Experiment Detail	CMMQA					Avg
	MMLU	MedQA	MedMCQA	PubMedQA	BioASQ	
MCA-RAG	78.26	74.42	75.27	75.00	81.71	76.74
MCA-RAG[no DCRMR]	73.91	70.93	72.04	71.67	76.83	72.97
Only SCREP	69.57	66.28	65.59	68.33	75.61	68.02
Only LLM+COT	60.87	65.12	65.59	51.67	67.07	63.08

TABLE 6.1: Effect of criteria in MCA-RAG. MCA-RAG[no DCRMR] refers to not using complexity-awareness, and the latter two refer to not using any criterion

Utilizing all criteria to select the most suitable strategy from 3 options (DCRMR, SCREP, or no retrieval) for LLMs outperforms other configurations on all datasets (Table 6.1). Even when complexity-awareness is absent, the selective application of SCREP or no

retrieval based on self-awareness and knowledge-awareness still provides significant benefits over a one-size-fits-all retrieval approach, suggesting that retrieval can occasionally worsen the response. The consistent performance drop when criteria are removed suggests that each criterion of the MCA-RAG is crucial for applying the appropriate strategy to optimize performance across diverse queries. Moreover, we find complex retrieval strategies cause higher latency (Table 6.2) while they generally enhance response accuracy. Therefore, MCA-RAG is also important for CPQA’s efficiency.

Predict_labels	Time/Query (Sec.)	Percentage
No Retrieval	2.41	33.14%
SCREP	24.09	34.88%
DCRMR	147.82	31.98%

TABLE 6.2: The average time per query and the percentage of the predicted labels from the classifiers over all samples

Chapter 7

Conclusion

We conducted experiments to evaluate the impact of SCREP on obtaining the right information for RAG. By expanding our knowledge sources beyond PubMed and PMC to include clinical guidelines (e.g. NLM books) and textbooks used for USMLE preparation, we ensured a broad spectrum of relevant information. Our hybrid search provides a more **comprehensive retrieval**. Hier-EP is *the first* approach to integrate a comprehensive filtering mechanism with adaptive granularity control. We ensure that **high-quality information** is selected by K-means clustering with multiple quality factors (including relevance, length, publication year, citations, and data source) whose weights are automatically tuned to optimize the relative importance of each factor, while adaptive granularity control minimizes distractions by focusing on the most relevant content. This is particularly critical in oncology, where the comprehensiveness, relevance, reliability, and timeliness of information are necessary for precision cancer care [123] and making decisions about complex treatment [124]. Moreover, cancer-treatment-related queries require sentence-level evidence due to the detailed nature of treatment protocols while other cancer-related queries may need paragraph-level information for overall consideration, and our adaptive granularity control in Hier-EP algorithm excels in this regard.

Our experiments on different datasets (CMMQA and CMCQA) demonstrate the effectiveness of SCREP in improving the performance of multiple LLMs while maintaining efficiency. The results indicate that 1) queries exhibit distinct source preferences, with NLM books offering more relevant evidence for clinical queries, 2) the use of Hier-EP, which considers multiple quality factors and adaptively adjusts the granularity of the text passed to the prompt, enhances performance and saves time, 3) queries have different preferences for Hier-EP parameters, with PubMed queries requiring more sentence-level

evidence, and 4) SCREP’s generalization ability significantly enhances LLMs performance across different datasets, albeit at the cost of increased response time. Overall, llama-3-8b is the **right LLM** considering all factors. Notably, the improvement in self-awareness accuracy is attributed to RAG enhancing the LLM’s understanding of its knowledge boundaries, while the improvement in reflection accuracy is primarily due to RAG increasing response accuracy. However, using LLM self-awareness and reflection for negative rejection remains unreliable. Moreover, our Comprehensibility and Verifiability Booster (CVB) incorporates source attributes, citations, readability adjustment, fluency optimization, and medical terminology simplification. Our qualitative analysis of the generated output **in right form** reveals discrepancies between the “step-by-step thinking” part and the “Citations” part.

Our novel Multi-Criteria Adaptive RAG (MCA-RAG) uses three essential orthogonal criteria (Figure 7.1) to decide the most suitable strategy (LLM+COT, SCREP, or DCRMR using Hier-EP per step) for retrieval **right information**. Decisions are made through few-shot prompt technology, plug-and-play classifiers, or both, allowing adaptive retrieval sequence judgment without compromising model performance. DCRMR, a novel Dual-LLM Collaborative React-based Multi-Step Retrieval method, leverages the strengths of two LLMs: LLaMA-3-8B generates per-step responses, while LLaMA-3-70B functions as the ReAct agent, handling complex reasoning and decision-making tasks such as planning, routing, and reflection. Using the CMMQA dataset, we evaluated MCA-RAG’s retrieval performance and efficiency when LLaMA-3-8B is the CPQA generator. The results demonstrate MCA-RAG’s ability to balance response accuracy (e.g., improving LLaMA-3-8B’s average accuracy beyond GPT-4) and efficiency (e.g., significantly reducing retrieval time by selecting the most suitable approach).

	FLARE	AdaptiveRAG SKR	Self-RAG	UAR	MCA-RAG
	[82]	[31]	[83]	[55]	Our work
Knowledge-awareness	N	N	N	N	Y
Self-awareness	Y	N	Y	Y	Y
Complexity-awareness	N	Y	N	N	Y
Time Awareness	N	N	N	N	N
Intent Awareness	N	N	N	N	N

TABLE 7.1: Criteria in MCA-RAG are carefully selected (e.g., Time Awareness and intent awareness are excluded due to **domain**-specific considerations in section 6.2)

For constructing evaluation datasets, we utilized the MeSH-filter method. However, it might not catch indirectly related queries, such as containing symptoms associated with cancer but not explicitly mentioning terms under the “neoplasm” MeSH sub-tree. Future work could implement more sophisticated filtering methods using biomedical Named Entity Recognition (NER) and relation extraction to identify entities and relationships

specific to cancer. For the NER and reasoning problem with RAG, Langchain and LangGraph will also be good tools to explore. Moreover, we acknowledge Multiple-choice QA Dataset limitations in capturing the nuanced differences between generated text answers and human-created long text answers [125].

We created a cancer-related consumer QA dataset (Figure 7.1) from the LiveQA-Med Training set2 [126]. However, the count of cancer-related queries in this dataset is smaller than in multiple-choice datasets (CMMQA:344, CMCQA:76). Small sample sizes increase variance and reduce the reliability of the conclusions. For accurate experimentation, we need to construct a larger and more representative dataset to mitigate bias and improve the validity of the results.

Another important thing for evaluation is human feedback mechanism, specifically targeting difficult-to-quantify but crucial metrics for cancer patients, such as answer relevance, professionalism, comprehensibility, empathy, and practicality. By intuitive “agree” or “disagree”, users can evaluate answers easily. If users have completed feedback on all metrics, two new answers based on feedback will be generated and CPQA system will ask the user to select a better one. By recording user clicks on the ratings and selections of regenerated answers, we can collect valuable data for subsequent reinforcement learning. This approach aligns with reinforcement learning from human feedback (RLHF) which has been shown to be effective in fine-tuning models to better align with human preferences [127].



FIGURE 7.1: Cancer-related text consumer QA dataset (CTCQA)

In SCREP, data analysis determines the proportion of documents retrieved from different sources. Although this method proved more effective overall than equal-quantity retrieval, it may not be the optimal retrieval quantity ratio for a specific query. Section 4.3.1 shows that different datasets prefer specific data sources. We should develop a method to adjust the retrieval quantity ratio adaptively in hybrid search. Our Hier-EP algorithm aimed to minimize noise that can cause faults in RAG systems [128]. However, recent studies find that some noise might improve generation quality [129]. The impact of noise remains unclear, and future research should investigate how to classify noise and selectively use noise to enhance retrieval performance in the Hier-EP algorithm. Another future research direction is to incorporate more quality factors, such as

the impact factor of journals. Moreover, the relative popularity in the cancer subfield can be seen as new features affecting the time decay in the Hier-EP algorithm.

Due to time and computational resource constraints, our MCA-RAG did not fine-tune the LLM model on the medical literature corpus to enhance the generator LLM’s self-awareness and knowledge-awareness. This method could improve the RAG system’s self-awareness and knowledge-awareness and warrants exploration. Additionally, as a React-based retrieval, we could consider using DSPY’s BootstrapFewShotWithRandomSearch starting from an earlier bootstrap of the ReAct program, which has been proven to outperform expert human reasoning [108].

Appendix A

Appendix

A.1 A glossary of acronyms

A glossary of acronyms

Acronym	Full Name
RAG	Retrieval-Augmented Generation
CPQA	Cancer Patient Question Answering System
PMC	PubMed Central
NCBI	National Center for Biotechnology Information
SCREP	Single-Step Comprehensive Retrieval With Evidence-Processing
CMMQA	Cancer-related Multiple-choice Medical QA Dataset
CVB	Comprehensibility and Verifiability Booster
MCA-RAG	Multi-Criteria Adaptive RAG
DCRMR	Dual-LLM Collaborative React-based Multi-Step Retrieval
Hier-EP	Hierarchical Evidence-Processing
HSRT	Hybrid Semantic Real-time document retrieval
RRF	Reciprocal Rank Fusion
SEOS	Semantic Enhanced Overlap Segmentation
CoT	Chain-of-Thought
MeSH	Medical Subject Headings
USMLE	US Medical Licensing Examination
MMLU	Massive Multitask Language Understanding
SUP	whether all statements of answers can be supported by evidence
USE	whether all statements of answers are a useful response to the question

A.2 Introduction of Original Medical QA dataset

Original Medical QA dataset of CMMQA: PubMedQA [101] and BioASQ [43] are biomedical QA datasets with answers from biomedical research, i.e., the online NCBI source. MMLU-Med [96] is a QA subset of six biomedicine-related tasks in Massive Multitask Language Understanding (MMLU) [98]. MedQA [99] collected questions from the US Medical Licensing Examination (USMLE). For MedMCQA, which contains questions from Indian medical entrance exams [100], only the Dev set with the ground truth was utilized.

Original Medical QA dataset of CMCQA: HealthFC-2 provides fact-checked questions focusing on consumer health questions and common topics users search for health advice online [102] and TREC-Health that contains popular health questions from two TREC-shared challenges with incorrect search engine results for health information [80].

A.3 More details(prompt contents) in MCA-RAG

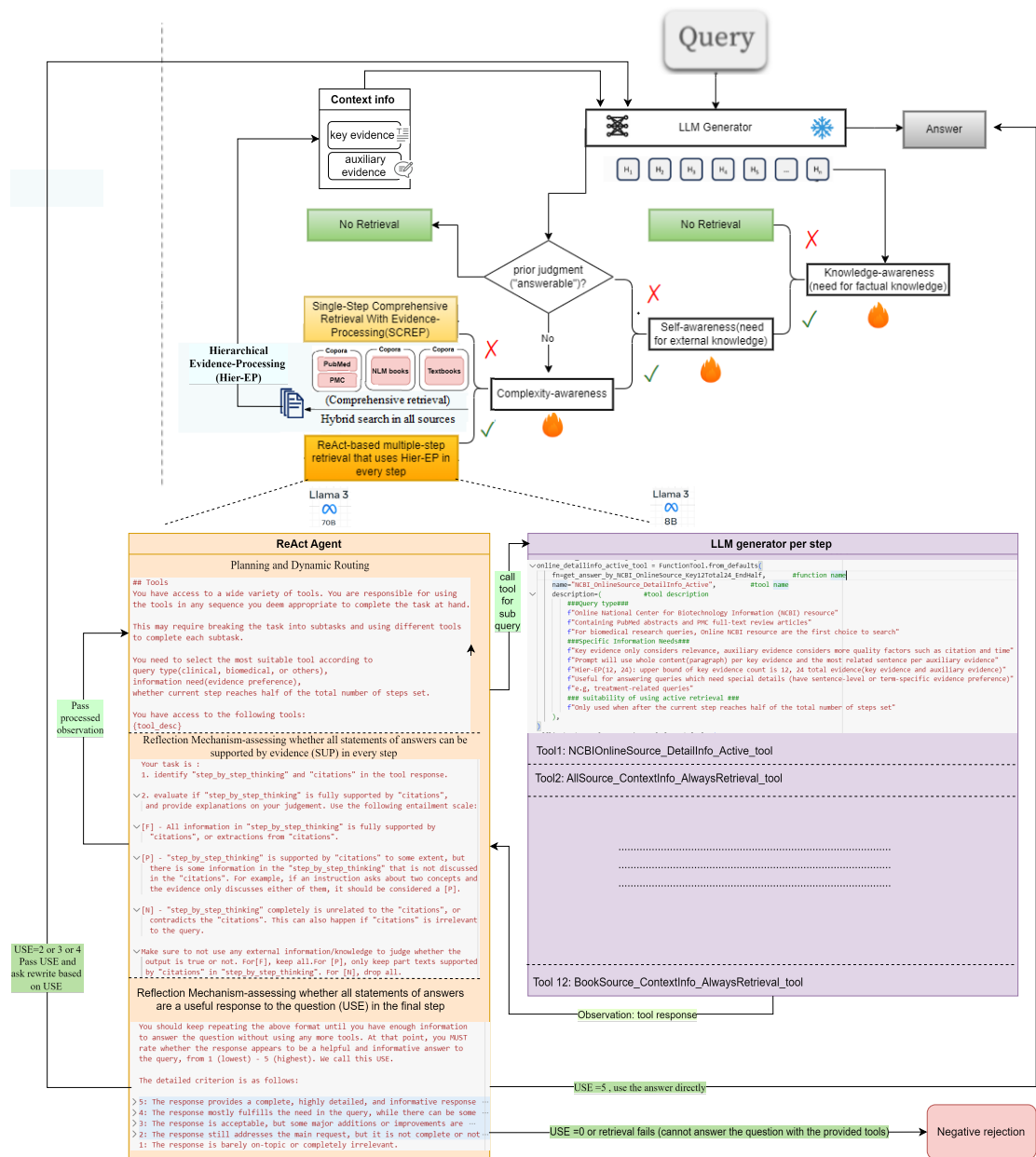


FIGURE A.1: The detail Framework of MCA-RAG, this is svg file, suggest zoom in

Bibliography

- [1] Ziwei Ji, Nayeon Lee, Rita Frieske, Tiezheng Yu, Dan Su, Yan Xu, Etsuko Ishii, Ye Jin Bang, Andrea Madotto, and Pascale Fung. Survey of hallucination in natural language generation. ACM Computing Surveys, 55(12):1–38, 2023.
- [2] Yejin Bang, Samuel Cahyawijaya, Nayeon Lee, Wenliang Dai, Dan Su, Bryan Wilie, Holy Lovenia, Ziwei Ji, Tiezheng Yu, Willy Chung, et al. A multitask, multilingual, multimodal evaluation of chatgpt on reasoning, hallucination, and interactivity. arXiv preprint arXiv:2302.04023, 2023.
- [3] Gordon V Cormack, Charles LA Clarke, and Stefan Buettcher. Reciprocal rank fusion outperforms condorcet and individual rank learning methods. In Proceedings of the 32nd international ACM SIGIR conference on Research and development in information retrieval, pages 758–759, 2009.
- [4] JuraJ Vladika and Florian Matthes. Improving health question answering with reliable and time-aware evidence retrieval. In Findings of the Association for Computational Linguistics: NAACL 2024, pages 4752–4763, 2024.
- [5] Benedikt Perak, Slobodan Beliga, and Ana Meštrović. Incorporating dialect understanding into llm using rag and prompt engineering techniques for causal commonsense reasoning. In Proceedings of the Eleventh Workshop on NLP for Similar Languages, Varieties, and Dialects (VarDial 2024), pages 220–229, 2024.
- [6] Valentin Liévin, Christoffer Egeberg Hother, Andreas Geert Motzfeldt, and Ole Winther. Can large language models reason about medical questions? Patterns, 2023.
- [7] Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altschmidt, Sam Altman, Shyamal Anadkat, et al. Gpt-4 technical report. arXiv preprint arXiv:2303.08774, 2023.

- [8] Machel Reid, Nikolay Savinov, Denis Teplyashin, Dmitry Lepikhin, Timothy Lill-icrap, Jean-baptiste Alayrac, Radu Soricut, Angeliki Lazaridou, Orhan Firat, Julian Schrittwieser, et al. Gemini 1.5: Unlocking multimodal understanding across millions of tokens of context. arXiv preprint arXiv:2403.05530, 2024.
- [9] AI Anthropic. The claude 3 model family: Opus, sonnet, haiku. Claude-3 Model Card, 2024.
- [10] Albert Q Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Deven-dra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guil-laume Lample, Lucile Saulnier, et al. Mistral 7b. arXiv preprint arXiv:2310.06825, 2023.
- [11] Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, et al. Llama: Open and efficient foundation language models. arXiv preprint arXiv:2302.13971, 2023.
- [12] Partha Pratim Ray. Chatgpt: A comprehensive review on background, applica-tions, key challenges, bias, ethics, limitations and future scope. Internet of Things and Cyber-Physical Systems, 2023.
- [13] Bhaskarjit Sarmah, Dhagash Mehta, Stefano Pasquali, and Tianjie Zhu. Towards reducing hallucination in extracting information from financial reports using large language models. In Proceedings of the Third International Conference on AI-ML Systems, pages 1–5, 2023.
- [14] Ziwei Ji, Tiezheng Yu, Yan Xu, Nayeon Lee, Etsuko Ishii, and Pascale Fung. To-wards mitigating llm hallucination via self reflection. In Findings of the Association for Computational Linguistics: EMNLP 2023, pages 1827–1843, 2023.
- [15] Daniel Truhn, Jorge S Reis-Filho, and Jakob Nikolas Kather. Large language models should be used as scientific reasoning engines, not knowledge databases. Nature medicine, 29(12):2983–2984, 2023.
- [16] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, et al. Retrieval-augmented generation for knowledge-intensive nlp tasks. Advances in Neural Information Processing Systems, 33:9459–9474, 2020.
- [17] Kelvin Guu, Kenton Lee, Zora Tung, Panupong Pasupat, and Mingwei Chang. Retrieval augmented language model pre-training. In International conference on machine learning, pages 3929–3938. PMLR, 2020.

- [18] Minki Kang, Seanie Lee, Jinheon Baek, Kenji Kawaguchi, and Sung Ju Hwang. Knowledge-augmented reasoning distillation for small language models in knowledge-intensive tasks. Advances in Neural Information Processing Systems, 36, 2024.
- [19] David Soong, Sriram Sridhar, Han Si, Jan-Samuel Wagner, Ana Caroline Costa Sá, Christina Y Yu, Kubra Karagoz, Meijian Guan, Hisham Hamadeh, and Brandon W Higgs. Improving accuracy of gpt-3/4 results on biomedical data using a retrieval-augmented language model. arXiv preprint arXiv:2305.17116, 2023.
- [20] Cyril Zakka, Rohan Shad, Akash Chaurasia, Alex R Dalal, Jennifer L Kim, Michael Moor, Robyn Fong, Curran Phillips, Kevin Alexander, Euan Ashley, et al. Almanac—retrieval-augmented language models for clinical medicine. NEJM AI, 1(2):AIoa2300068, 2024.
- [21] Aakanksha Naik, Sravanthi Parasa, Sergey Feldman, Lucy Lu Wang, and Tom Hope. Literature-augmented clinical outcome prediction. In 2022 Findings of the Association for Computational Linguistics: NAACL 2022, pages 438–453. Association for Computational Linguistics (ACL), 2022.
- [22] Giacomo Frisoni, Miki Mizutani, Gianluca Moro, and Lorenzo Valgimigli. Bioreader: a retrieval-enhanced text-to-text transformer for biomedical literature. In Proceedings of the 2022 conference on empirical methods in natural language processing, pages 5770–5793, 2022.
- [23] Qiao Jin, Robert Leaman, and Zhiyong Lu. Pubmed and beyond: biomedical literature search in the age of artificial intelligence. Ebiomedicine, 100, 2024.
- [24] Chaoyi Wu, Weixiong Lin, Xiaoman Zhang, Ya Zhang, Weidi Xie, and Yanfeng Wang. Pmc-llama: toward building open-source language models for medicine. Journal of the American Medical Informatics Association, page ocae045, 2024.
- [25] Eric W Sayers, Jeff Beck, Evan E Bolton, J Rodney Brister, Jessica Chan, Donald C Comeau, Ryan Connor, Michael DiCuccio, Catherine M Farrell, Michael Feldgarden, et al. Database resources of the national center for biotechnology information. Nucleic Acids Research, 52(D1):D33, 2024.
- [26] Majid Moshirfar, Amal W Altaf, Isabella M Stoakes, Jared J Tuttle, and Phillip C Hoopes. Artificial intelligence in ophthalmology: a comparative analysis of gpt-3.5, gpt-4, and human expertise in answering statpearls questions. Cureus, 15(6), 2023.
- [27] Daniel Gillick, Alessandro Presta, and Gaurav Singh Tomar. End-to-end retrieval in continuous space. arXiv preprint arXiv:1811.08008, 2018.

- [28] Vladimir Karpukhin, Barlas Oğuz, Sewon Min, Patrick Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen-tau Yih. Dense passage retrieval for open-domain question answering. arXiv preprint arXiv:2004.04906, 2020.
- [29] Nandan Thakur, Nils Reimers, Andreas Rücklé, Abhishek Srivastava, and Iryna Gurevych. Beir: A heterogenous benchmark for zero-shot evaluation of information retrieval models. arXiv preprint arXiv:2104.08663, 2021.
- [30] Stephen Robertson, Hugo Zaragoza, et al. The probabilistic relevance framework: Bm25 and beyond. Foundations and Trends® in Information Retrieval, 3(4): 333–389, 2009.
- [31] Soyeong Jeong, Jinheon Baek, Sukmin Cho, Sung Ju Hwang, and Jong C Park. Adaptive-rag: Learning to adapt retrieval-augmented large language models through question complexity. In Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers), pages 7029–7043, 2024.
- [32] Minbyul Jeong, Jiwoong Sohn, Mujeen Sung, and Jaewoo Kang. Improving medical reasoning through retrieval and self-reflection with retrieval-augmented large language models. arXiv preprint arXiv:2401.15269, 2024.
- [33] Nelson F Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. Lost in the middle: How language models use long contexts. Transactions of the Association for Computational Linguistics, 12: 157–173, 2024.
- [34] Thomas Merth, Qichen Fu, Mohammad Rastegari, and Mahyar Najibi. Superposition prompting: Improving and accelerating retrieval-augmented generation. arXiv e-prints, pages arXiv-2404, 2024.
- [35] Moshe Berchansky, Peter Izsak, Avi Caciularu, Ido Dagan, and Moshe Wasserblat. Optimizing retrieval-augmented reader models via token elimination. arXiv preprint arXiv:2310.13682, 2023.
- [36] Zhiruo Wang, Jun Araki, Zhengbao Jiang, Md Rizwan Parvez, and Graham Neubig. Learning to filter context for retrieval-augmented generation. arXiv preprint arXiv:2311.08377, 2023.
- [37] Emma C Scott, Andrea C Baines, Yutao Gong, Rodney Moore Jr, Gulsum E Pamuk, Haleh Saber, Ashim Subedee, Matthew D Thompson, Wenming Xiao, Richard Pazdur, et al. Trends in the approval of cancer therapies by the fda in the twenty-first century. Nature Reviews Drug Discovery, 22(8):625–640, 2023.

- [38] Constantino Carlos Reyes-Aldasoro. The proportion of cancer-related entries in pubmed has increased considerably; is cancer truly “the emperor of all maladies”? PloS one, 12(3):e0173671, 2017.
- [39] Jin Ge, Steve Sun, Joseph Owens, Victor Galvez, Oksana Gologorskaya, Jennifer C Lai, Mark J Pletcher, and Ki Lai. Development of a liver disease-specific large language model chat interface using retrieval augmented generation. Hepatology, pages 10–1097, 2024.
- [40] Alejandro Lozano, Scott L Fleming, Chia-Chun Chiang, and Nigam Shah. Clinfo. ai: An open-source retrieval-augmented large language model system for answering medical questions using scientific literature. In PACIFIC SYMPOSIUM ON BIOCOMPUTING 2024, pages 8–23. World Scientific, 2023.
- [41] Giovanni Maria Iannantuono, Dara Bracken-Clarke, Charalampos S Floudas, Mario Roselli, James L Gulley, and Fatima Karzai. Applications of large language models in cancer care: current evidence and future perspectives. Frontiers in Oncology, 13:1268915, 2023.
- [42] Malik Sallam. Chatgpt utility in healthcare education, research, and practice: systematic review on the promising perspectives and valid concerns. In Healthcare, volume 11, page 887. MDPI, 2023.
- [43] Guangzhi Xiong, Qiao Jin, Zhiyong Lu, and Aidong Zhang. Benchmarking retrieval-augmented generation for medicine. arXiv e-prints, pages arXiv–2402, 2024.
- [44] Yizheng Huang and Jimmy Huang. A survey on retrieval-augmented text generation for large language models. arXiv preprint arXiv:2404.10981, 2024.
- [45] Amro Khasawneh, Ian Kratzke, Karthik Adapa, Lawrence Marks, and Lukasz Mazur. Effect of notes’ access and complexity on opennotes’ utility. Applied Clinical Informatics, 13(05):1015–1023, 2022.
- [46] Halil Kilicoglu, Asma Ben Abacha, Yassine M’rabet, Kirk Roberts, Laritza Rodriguez, Sonya Shooshan, and Dina Demner-Fushman. Annotating named entities in consumer health questions. In Proceedings of the Tenth International Conference on Language Resources and Evaluation (LREC’16), pages 3325–3332, 2016.
- [47] Halil Kilicoglu, Asma Ben Abacha, Yassine Mrabet, Sonya E Shooshan, Laritza Rodriguez, Kate Masterton, and Dina Demner-Fushman. Semantic annotation of consumer health questions. BMC bioinformatics, 19:1–28, 2018.

- [48] Halil Kilicoglu, Marcelo Fiszman, Kirk Roberts, and Dina Demner-Fushman. An ensemble method for spelling correction in consumer health questions. In AMIA Annual Symposium Proceedings, volume 2015, page 727. American Medical Informatics Association, 2015.
- [49] Chris J Lu, Alan R Aronson, Sonya E Shooshan, and Dina Demner-Fushman. Spell checker for consumer language (cspell). Journal of the American Medical Informatics Association, 26(3):211–218, 2019.
- [50] Asma Ben Abacha, Eugene Agichtein, Yuval Pinter, and Dina Demner-Fushman. Overview of the medical question answering task at trec 2017 liveqa. In TREC, pages 1–12, 2017.
- [51] Asma Ben Abacha, Yassine Mrabet, Mark Sharp, Travis R Goodwin, Sonya E Shooshan, and Dina Demner-Fushman. Bridging the gap between consumers’ medication questions and trusted answers. In MEDINFO 2019: Health and Wellbeing e-Networks for All, pages 25–29. IOS Press, 2019.
- [52] DC MANNING. Introduction to information retrieval. Journal of the American Statistical Association, 15, 2008.
- [53] Michael Gusenbauer and Neal R Haddaway. Which academic search systems are suitable for systematic reviews or meta-analyses? evaluating retrieval qualities of google scholar, pubmed, and 26 other resources. Research synthesis methods, 11(2):181–217, 2020.
- [54] Jonathan Kans. Entrez direct: E-utilities on the unix command line. In Entrez programming utilities help [Internet]. National Center for Biotechnology Information (US), 2024.
- [55] Akari Asai, Zeqiu Wu, Yizhong Wang, Avirup Sil, and Hannaneh Hajishirzi. Self-rag: Learning to retrieve, generate, and critique through self-reflection. In The Twelfth International Conference on Learning Representations, 2023.
- [56] Alex Mallen, Akari Asai, Victor Zhong, Rajarshi Das, Daniel Khashabi, and Hannaneh Hajishirzi. When not to trust language models: Investigating effectiveness of parametric and non-parametric memories. In Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), pages 9802–9822, 2023.
- [57] Irena Spasić, Jacqueline Livsey, John A Keane, and Goran Nenadić. Text mining of cancer-related information: review of current status and future directions. International journal of medical informatics, 83(9):605–623, 2014.

- [58] Harsh Trivedi, Niranjan Balasubramanian, Tushar Khot, and Ashish Sabharwal. Interleaving retrieval with chain-of-thought reasoning for knowledge-intensive multi-step questions. arXiv preprint arXiv:2212.10509, 2022.
- [59] Ori Yoran, Tomer Wolfson, Ori Ram, and Jonathan Berant. Making retrieval-augmented language models robust to irrelevant context. In The Twelfth International Conference on Learning Representations, 2023.
- [60] Jeffrey M Weinfeld and Kate Finkelstein. How to answer your clinical questions more efficiently. Family practice management, 12(7):37–41, 2005.
- [61] Yubo Wang, Xueguang Ma, and Wenhui Chen. Augmenting black-box llms with medical textbooks for clinical question answering. arXiv preprint arXiv:2309.02233, 2023.
- [62] Lucinda Wynter, Annette Burgess, Eszter Kalman, Jack Edward Heron, and Jane Bleasel. Medical students: what educational resources are they using? BMC medical education, 19:1–8, 2019.
- [63] Marco Wrzalik and Dirk Krechel. Cort: Complementary rankings from transformers. In Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, pages 4194–4204, 2021.
- [64] Yi Luan, Jacob Eisenstein, Kristina Toutanova, and Michael Collins. Sparse, dense, and attentional representations for text retrieval. Transactions of the Association for Computational Linguistics, 9:329–345, 2021.
- [65] Negar Arabzadeh, Xinyi Yan, and Charles LA Clarke. Predicting efficiency/effectiveness trade-offs for dense vs. sparse retrieval strategy selection. In Proceedings of the 30th ACM International Conference on Information & Knowledge Management, pages 2862–2866, 2021.
- [66] Xiaoqian Li, Ercong Nie, and Sheng Liang. From classification to generation: Insights into crosslingual retrieval augmented icl. arXiv preprint arXiv:2311.06595, 2023.
- [67] Huayang Li, Yixuan Su, Deng Cai, Yan Wang, and Lemao Liu. A survey on retrieval-augmented text generation. arXiv preprint arXiv:2202.01110, 2022.
- [68] Penghao Zhao, Hailin Zhang, Qinhan Yu, Zhengren Wang, Yunteng Geng, Fangcheng Fu, Ling Yang, Wentao Zhang, and Bin Cui. Retrieval-augmented generation for ai-generated content: A survey. arXiv preprint arXiv:2402.19473, 2024.

- [69] Jerry Liu. Llamaindex. Acceso el, 6, 2022.
- [70] Wenhao Yu, Hongming Zhang, Xiaoman Pan, Kaixin Ma, Hongwei Wang, and Dong Yu. Chain-of-note: Enhancing robustness in retrieval-augmented language models. arXiv preprint arXiv:2311.09210, 2023.
- [71] Freda Shi, Xinyun Chen, Kanishka Misra, Nathan Scales, David Dohan, Ed H Chi, Nathanael Schärli, and Denny Zhou. Large language models can be easily distracted by irrelevant context. In International Conference on Machine Learning, pages 31210–31227. PMLR, 2023.
- [72] Fangyuan Xu, Weijia Shi, and Eunsol Choi. Recomp: Improving retrieval-augmented lms with compression and selective augmentation. arXiv preprint arXiv:2310.04408, 2023.
- [73] Sebastian Hofstätter, Jiecao Chen, Karthik Raman, and Hamed Zamani. Fid-light: Efficient and effective retrieval-augmented text generation. In Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval, pages 1437–1447, 2023.
- [74] Renat Aksitov, Chung-Ching Chang, David Reitter, Siamak Shakeri, and Yunhsuan Sung. Characterizing attribution and fluency tradeoffs for retrieval-augmented large language models. arXiv preprint arXiv:2302.05578, 2023.
- [75] Bernhard Kratzwald and Stefan Feuerriegel. Adaptive document retrieval for deep question answering. In Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing, pages 576–581, 2018.
- [76] Jack B Longwell, Ian Hirsch, Fernando Binder, Galileo Arturo Gonzalez Conchas, Daniel Mau, Raymond Jang, Rahul G Krishnan, and Robert C Grant. Performance of large language models on medical oncology examination questions. JAMA Network Open, 7(6), 2024.
- [77] Logesh Kumar Umapathi, Ankit Pal, and Malaikannan Sankarasubbu. Med-halt: Medical domain hallucination test for large language models. arXiv preprint arXiv:2307.15343, 2023.
- [78] Shubo Tian, Qiao Jin, Lana Yeganova, Po-Ting Lai, Qingqing Zhu, Xiuying Chen, Yifan Yang, Qingyu Chen, Won Kim, Donald C Comeau, et al. Opportunities and challenges for chatgpt and large language models in biomedicine and health. Briefings in Bioinformatics, 25(1):bbad493, 2024.

- [79] Asad J Choudhry, Yaser MK Baghdadi, Amy E Wagie, Elizabeth B Habermann, Stephanie F Heller, Donald H Jenkins, Daniel C Cullinane, and Martin D Zielinski. Readability of discharge summaries: with what level of information are we dismissing our patients? The American Journal of Surgery, 211(3):631–636, 2016.
- [80] Alexander Pugachev, Ekaterina Artemova, Alexander Bondarenko, and Pavel Braslavski. Consumer health question answering using off-the-shelf components. In European Conference on Information Retrieval, pages 571–579. Springer, 2023.
- [81] Jae QJ Liu, Kelvin TK Hui, Fadi Al Zoubi, Zing ZX Zhou, Dino Samartzis, Curtis CH Yu, Jeremy R Chang, and Arnold YL Wong. The great detectives: humans versus ai detectors in catching large language model-generated medical writing. International Journal for Educational Integrity, 20(1):8, 2024.
- [82] Zhengbao Jiang, Frank F Xu, Luyu Gao, Zhiqing Sun, Qian Liu, Jane Dwivedi-Yu, Yiming Yang, Jamie Callan, and Graham Neubig. Active retrieval augmented generation. In The 2023 Conference on Empirical Methods in Natural Language Processing, 2023.
- [83] Yile Wang, Peng Li, Maosong Sun, and Yang Liu. Self-knowledge guided retrieval augmentation for large language models. In Findings of the Association for Computational Linguistics: EMNLP 2023, pages 10303–10315, 2023.
- [84] Matthias Sauter, Elena Belousova, Mirjana P Benedik, Tom Carter, Vincent Cottin, Paolo Curatolo, Maria Dahlin, Lisa D’Amato, Guillaume B d’Augères, Petrus J de Vries, et al. Rare manifestations and malignancies in tuberous sclerosis complex: findings from the tuberous sclerosis registry to increase disease awareness (tosca). Orphanet journal of rare diseases, 16:1–15, 2021.
- [85] Retrieval-augmented generation for large language models: A survey.
- [86] Xingxuan Li, Ruochen Zhao, Yew Ken Chia, Bosheng Ding, Lidong Bing, Shafiq Joty, and Soujanya Poria. Chain of knowledge: A framework for grounding large language models with structured knowledge bases. arXiv preprint arXiv:2305.13269, 2023.
- [87] Qiao Jin, Won Kim, Qingyu Chen, Donald C Comeau, Lana Yeganova, W John Wilbur, and Zhiyong Lu. Medcpt: Contrastive pre-trained transformers with large-scale pubmed search logs for zero-shot biomedical information retrieval. Bioinformatics, 39(11):btad651, 2023.
- [88] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc V Le, Denny Zhou, et al. Chain-of-thought prompting elicits reasoning in

- large language models. Advances in neural information processing systems, 35: 24824–24837, 2022.
- [89] Zhihong Shao, Yeyun Gong, Yelong Shen, Minlie Huang, Nan Duan, and Weizhu Chen. Enhancing retrieval-augmented large language models with iterative retrieval-generation synergy. In Findings of the Association for Computational Linguistics: EMNLP 2023, pages 9248–9274, 2023.
- [90] Gangwoo Kim, Sungdong Kim, Byeongguk Jeon, Joonsuk Park, and Jaewoo Kang. Tree of clarifications: Answering ambiguous questions with retrieval-augmented large language models. arXiv preprint arXiv:2310.14696, 2023.
- [91] Yixuan Tang and Yi Yang. Multihop-rag: Benchmarking retrieval-augmented generation for multi-hop queries. arXiv e-prints, pages arXiv–2401, 2024.
- [92] Sun Kim, Lana Yeganova, and W John Wilbur. Meshable: searching pubmed abstracts by utilizing mesh and mesh-derived topical terms. Bioinformatics, 32(19):3044–3046, 2016.
- [93] Harsha Nori, Nicholas King, Scott Mayer McKinney, Dean Carignan, and Eric Horvitz. Capabilities of gpt-4 on medical challenge problems. arXiv e-prints, pages arXiv–2303, 2023.
- [94] Jakub Lála, Odhran O’Donoghue, Aleksandar Shtedritski, Sam Cox, Samuel G Rodrigues, and Andrew D White. Paperqa: Retrieval-augmented generative agent for scientific research. arXiv preprint arXiv:2312.07559, 2023.
- [95] Harsha Nori, Yin Tat Lee, Sheng Zhang, Dean Carignan, Richard Edgar, Nicolo Fusi, Nicholas King, Jonathan Larson, Yuanzhi Li, Weishung Liu, et al. Can generalist foundation models outcompete special-purpose tuning? case study in medicine. Medicine, 84(88.3):77–3, 2023.
- [96] Karan Singhal, Shekoofeh Azizi, Tao Tu, S Sara Mahdavi, Jason Wei, Hyung Won Chung, Nathan Scales, Ajay Tanwani, Heather Cole-Lewis, Stephen Pfohl, et al. Large language models encode clinical knowledge. Nature, 620(7972):172–180, 2023.
- [97] Valentin Liévin, Christoffer Egeberg Hother, Andreas Geert Motzfeldt, and Ole Winther. Can large language models reason about medical questions? Patterns, 5(3), 2024.
- [98] Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. Measuring massive multitask language understanding. In International Conference on Learning Representations, 2020.

- [99] Di Jin, Eileen Pan, Nassim Oufattole, Weng Wei-Hung, Hanyi Fang, and Peter Szolovits. What disease does this patient have? a large-scale open domain question answering dataset from medical exams. Applied Sciences, 11(14):6421, 2021.
- [100] Ankit Pal, Logesh Kumar Umapathi, and Malaikannan Sankarasubbu. Medmcqa: A large-scale multi-subject multi-choice dataset for medical domain question answering. In Conference on health, inference, and learning, pages 248–260. PMLR, 2022.
- [101] Qiao Jin, Bhuwan Dhingra, Zhengping Liu, William Cohen, and Xinghua Lu. Pubmedqa: A dataset for biomedical research question answering. In Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP), pages 2567–2577, 2019.
- [102] JuraJ Vladika, Phillip Schneider, and Florian Matthes. Healthfc: A dataset of health claims for evidence-based medical fact-checking. arXiv preprint arXiv:2309.08503, 2023.
- [103] Hao Yu, Aoran Gan, Kai Zhang, Shiwei Tong, Qi Liu, and Zhaofeng Liu. Evaluation of retrieval-augmented generation: A survey. arXiv e-prints, pages arXiv–2405, 2024.
- [104] Julianna Czum and Shane Parr. Bias in foundation models: primum non nocere or caveat emptor? Radiology: Artificial Intelligence, 5(6):e230384, 2023.
- [105] Di Jin, Eileen Pan, Nassim Oufattole, Wei-Hung Weng, Hanyi Fang, and Peter Szolovits. What disease does this patient have? a large-scale open domain question answering dataset from medical exams. arXiv preprint arXiv:2009.13081, 2020.
- [106] Marcel Schwarz, Kathryn Chapman, and Bertram Häussler. Multilingual medical entity recognition and cross-lingual zero-shot linking with facebook ai similarity search. In IberLEF@ SEPLN, 2022.
- [107] Kristina P Sinaga and Miin-Shen Yang. Unsupervised k-means clustering algorithm. IEEE access, 8:80716–80727, 2020.
- [108] Omar Khattab, Arnav Singhvi, Paridhi Maheshwari, Zhiyuan Zhang, Keshav Santhanam, Sri Vardhamanan, Saiful Haq, Ashutosh Sharma, Thomas T Joshi, Hanna Moazam, et al. Dspy: Compiling declarative language model calls into self-improving pipelines. arXiv preprint arXiv:2310.03714, 2023.
- [109] Derya Soydaner. A comparison of optimization algorithms for deep learning. International Journal of Pattern Recognition and Artificial Intelligence, 34(13): 2052013, 2020.

- [110] Derya Soydaner. Attention mechanism in neural networks: where it comes and where it goes. Neural Computing and Applications, 34(16):13371–13385, 2022.
- [111] Jian Zhang, Xu Wang, Hongyu Zhang, Hailong Sun, and Xudong Liu. Retrieval-based neural source code summarization. In Proceedings of the ACM/IEEE 42nd International Conference on Software Engineering, pages 1385–1397, 2020.
- [112] Chi Yu, Guang Yang, Xiang Chen, Ke Liu, and Yanlin Zhou. Bashexplainer: Retrieval-augmented bash code comment generation based on fine-tuned codebert. In 2022 IEEE International Conference on Software Maintenance and Evolution (ICSME), pages 82–93. IEEE, 2022.
- [113] Lee Xiong, Chenyan Xiong, Ye Li, Kwok-Fung Tang, Jialin Liu, Paul Bennett, Junaid Ahmed, and Arnold Overwijk. Approximate nearest neighbor negative contrastive learning for dense text retrieval. arXiv preprint arXiv:2007.00808, 2020.
- [114] Adam EM Eltorai, Soha Ghanian, Charles A Adams Jr, Christopher T Born, and Alan H Daniels. Readability of patient education materials on the american association for surgery of trauma website. Archives of trauma research, 3(2), 2014.
- [115] Marina Solnyshkina, Radif Zamaletdinov, Ludmila Gorodetskaya, and Azat Gabitov. Evaluating text complexity and flesch-kincaid grade level. Journal of social studies education research, 8(3):238–248, 2017.
- [116] Ruiyang Ren, Yuhao Wang, Yingqi Qu, Wayne Xin Zhao, Jing Liu, Hao Tian, Hua Wu, Ji-Rong Wen, and Haifeng Wang. Investigating the factual knowledge boundary of large language models with retrieval augmentation. arXiv preprint arXiv:2307.11019, 2023.
- [117] Justin Kruger and David Dunning. Unskilled and unaware of it: how difficulties in recognizing one’s own incompetence lead to inflated self-assessments. Journal of personality and social psychology, 77(6):1121, 1999.
- [118] Aniket Kumar Singh, Suman Devkota, Bishal Lamichhane, Uttam Dhakal, and Chandra Dhakal. The confidence-competence gap in large language models: Acognitive study. arXiv preprint arXiv:2309.16145, 2023.
- [119] Qinyuan Cheng, Xiaonan Li, Shimin Li, Qin Zhu, Zhangyue Yin, Yunfan Shao, Linyang Li, Tianxiang Sun, Hang Yan, and Xipeng Qiu. Unified active retrieval for retrieval augmented generation. arXiv preprint arXiv:2406.12534, 2024.
- [120] Nicholas R Rydzewski, Deepak Dinakaran, Shuang G Zhao, Eytan Ruppín, Baris Turkbey, Deborah E Citrin, and Krishnan R Patel. Comparative evaluation of llms in clinical oncology. Nejm Ai, 1(5):AIoa2300151, 2024.

- [121] Qinyuan Cheng, Tianxiang Sun, Xiangyang Liu, Wenwei Zhang, Zhangyue Yin, Shimin Li, Linyang Li, Zhengfu He, Kai Chen, and Xipeng Qiu. Can ai assistants know what they don't know? In Forty-first International Conference on Machine Learning.
- [122] Zhaorui Yang, Tianyu Pang, Haozhe Feng, Han Wang, Wei Chen, Minfeng Zhu, and Qian Liu. Self-distillation bridges distribution gap in language model fine-tuning. arXiv preprint arXiv:2402.13669, 2024.
- [123] Maria Schwaederle, Melissa Zhao, J Jack Lee, Alexander M Eggermont, Richard L Schilsky, John Mendelsohn, Vladimir Lazar, and Razelle Kurzrock. Impact of precision medicine in diverse cancers: a meta-analysis of phase ii clinical trials. Journal of clinical oncology, 33(32):3817–3825, 2015.
- [124] Ibiayi Dagogo-Jack and Alice T Shaw. Tumour heterogeneity and resistance to cancer therapies. Nature reviews Clinical oncology, 15(2):81–94, 2018.
- [125] Dina Demner-Fushman, Yassine Mrabet, and Asma Ben Abacha. Consumer health information and question answering: helping consumers find answers to their health-related information needs. Journal of the American Medical Informatics Association, 27(2):194–201, 2020.
- [126] Asma Ben Abacha, Eugene Agichtein, Yuval Pinter, and Dina Demner-Fushman. Overview of the medical question answering task at trec 2017 liveqa. In TREC 2017, 2017.
- [127] Wei Xiong, Hanze Dong, Chenlu Ye, Ziqi Wang, Han Zhong, Heng Ji, Nan Jiang, and Tong Zhang. Iterative preference learning from human feedback: Bridging theory and practice for rlhf under kl-constraint. In Forty-first International Conference on Machine Learning, 2024.
- [128] Scott Barnett, Stefanus Kurniawan, Srikanth Thudumu, Zach Brannelly, and Mohamed Abdelrazek. Seven failure points when engineering a retrieval augmented generation system. In Proceedings of the IEEE/ACM 3rd International Conference on AI Engineering-Software Engineering for AI, pages 194–199, 2024.
- [129] Florin Cuconasu, Giovanni Trappolini, Federico Siciliano, Simone Filice, Cesare Campagnano, Yoelle Maarek, Nicola Tonellotto, and Fabrizio Silvestri. The power of noise: Redefining retrieval for rag systems. In Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval, pages 719–729, 2024.